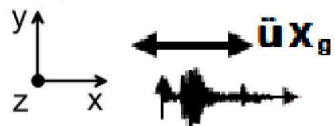
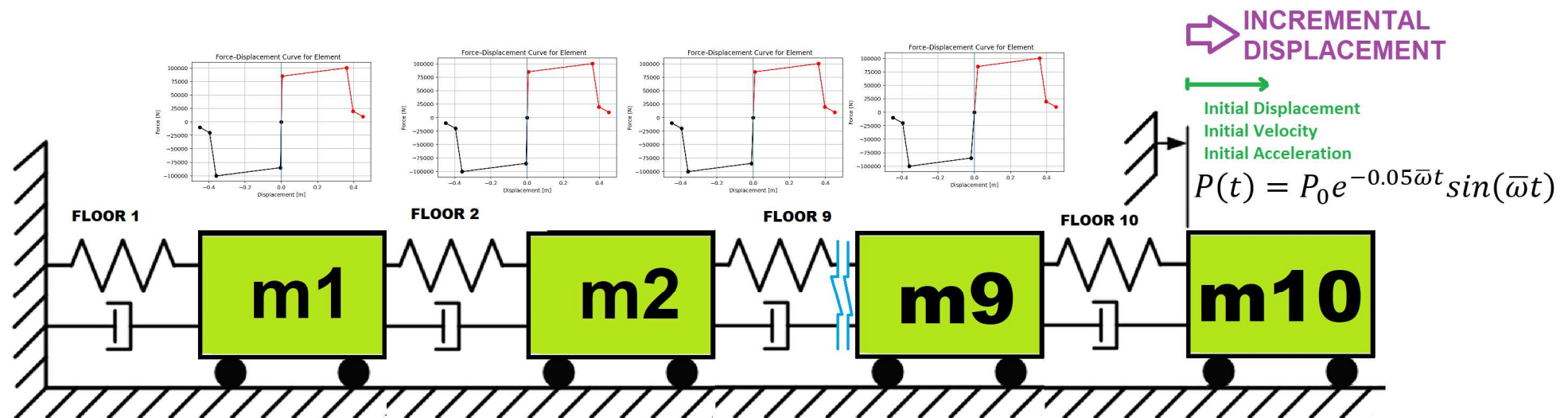


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

PUSHOVER ANALYSIS OF A MULTI-DEGREE-OF- FREEDOM STRUCTURE VIA EVALUATION OF A MULTILINEAR FITTING CURVE

THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

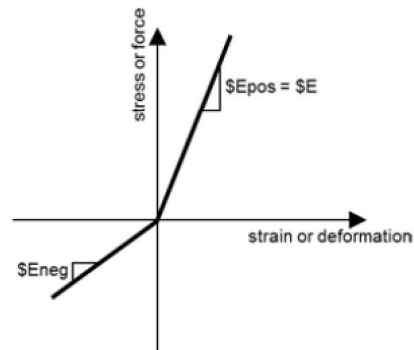
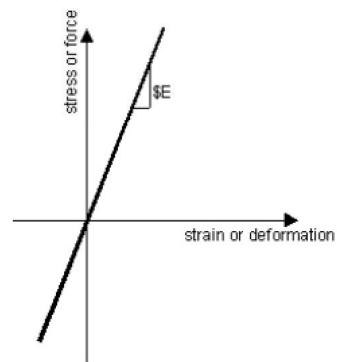
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

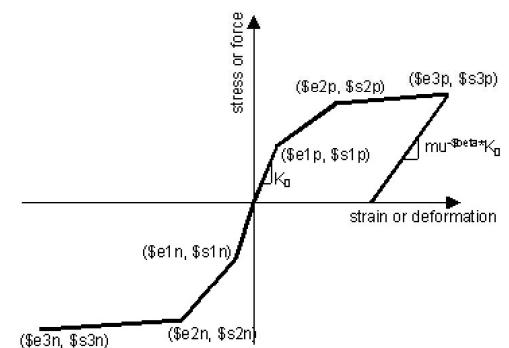
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

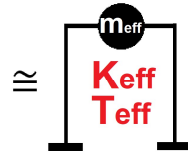
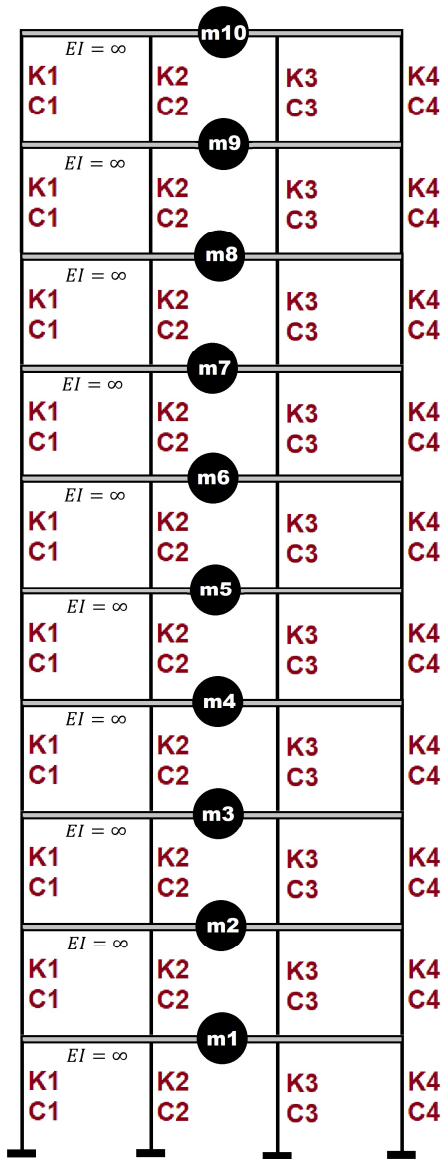
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



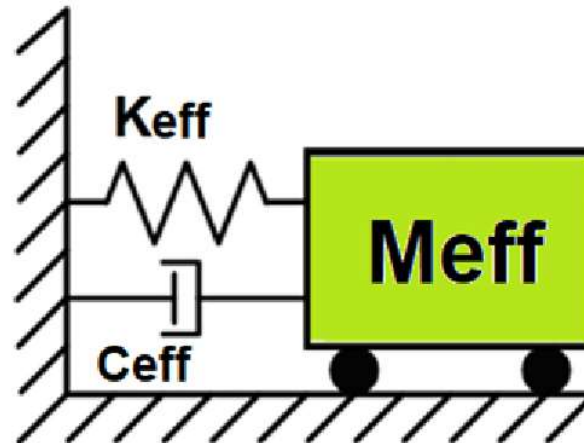
INELASTIC MATERIAL



Displacement-based seismic analysis transformation,
converting a multi-degree-of-freedom (MDOF) system
into an equivalent single-degree-of-freedom (SDOF)
system for seismic assessment

$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i} \quad m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d} \quad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

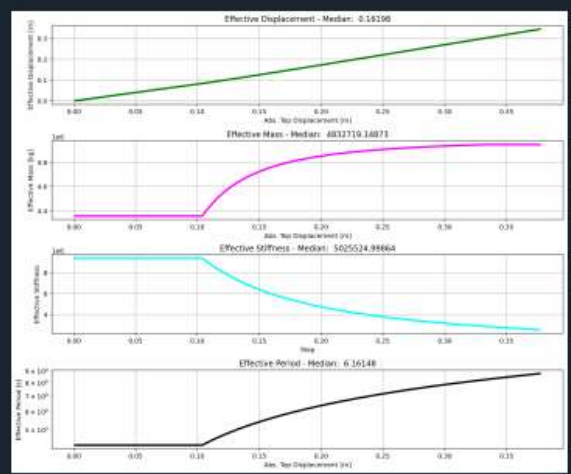
C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._PUSHOVER_MDOF+DUCT_SENSITIVITY_PUSHOVER_MDOF.py

DUCT_SENSITIVITY_PUSHOVER_MDOF.py

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # SENSITIVITY ANALYSIS OF COLUMN DUCTILITY RATIO AND OPENSEES VIA PUSHOVER ANALYSIS
4 # OF A MULTI-DEGREE-OF-FREEDOM STRUCTURE
5 # AND EVALUATION OF A MULTILINEAR FITTING CURVE FOR STRUCTURAL ELASTIC STIFFNESS,
6 # PLASTIC STIFFNESS, DUCTILITY RATIO, AND OVER-STRENGTH FACTOR
7 #-----
8 # EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANAL
9 #-----
10 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
11 # EMAIL: salar.d.ghashghaei@gmail.com
12 #####
13
14 # =====
15 # SENSITIVITY ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM (MDOF) BUILDING FRAME
16 #-----
17 # Effect of Column Ductility Ratio on the Equivalent SDOF System and on the
18 # Multilinear Idealization of the Pushover Curve
19 #
20 # METHODOLOGY
21 #-----
22 # 1. A 10-story MDOF shear-building frame is modelled in OpenSeesPy
23 # (10 lumped masses, 4 parallel zeroLength springs per floor).
24 #
25 # 2. The COLUMN DUCTILITY RATIO is chosen as the single design parameter and is
26 # swept over 21 equally spaced values between DUCT_MIN and DUCT_MAX.
27 #
28 # 3. For every value of the swept parameter, a nonlinear PUSHOVER ANALYSIS is
29 # carried out under displacement control applied at the top node.
30 #
31 # 4. The resulting base-shear / displacement curve is post-processed to derive the
32 # EQUIVALENT SDOF SYSTEM (effective displacement, mass, stiffness, and period)
33 # for displacement-based seismic design.
34 #
```

....VARIATION_MASS_OR_PERIOD\OPTIMIZATION_KE_PERIOD_VARIATIONS

21 %



IPython Console

Plots

Variable Explorer

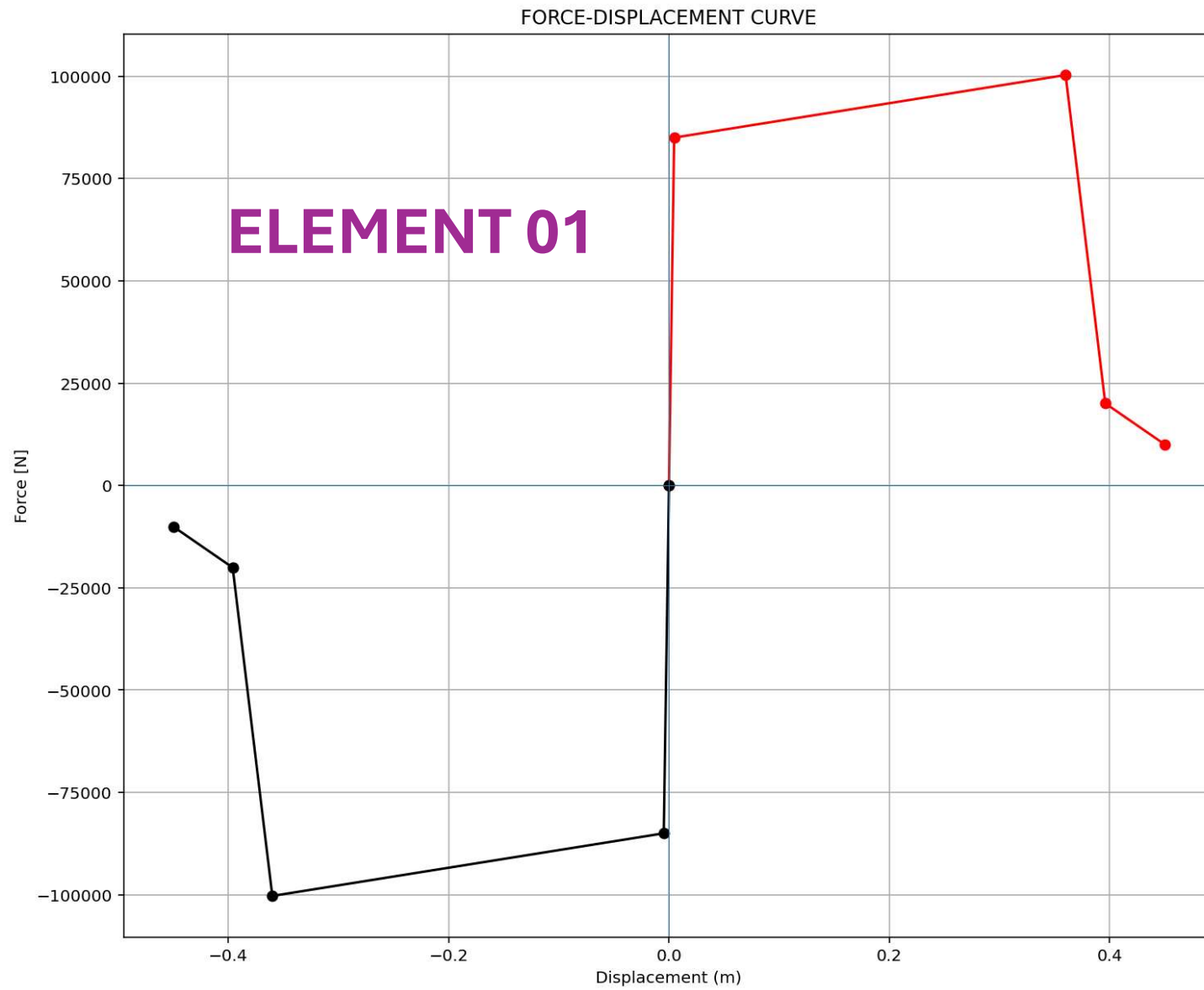
Help

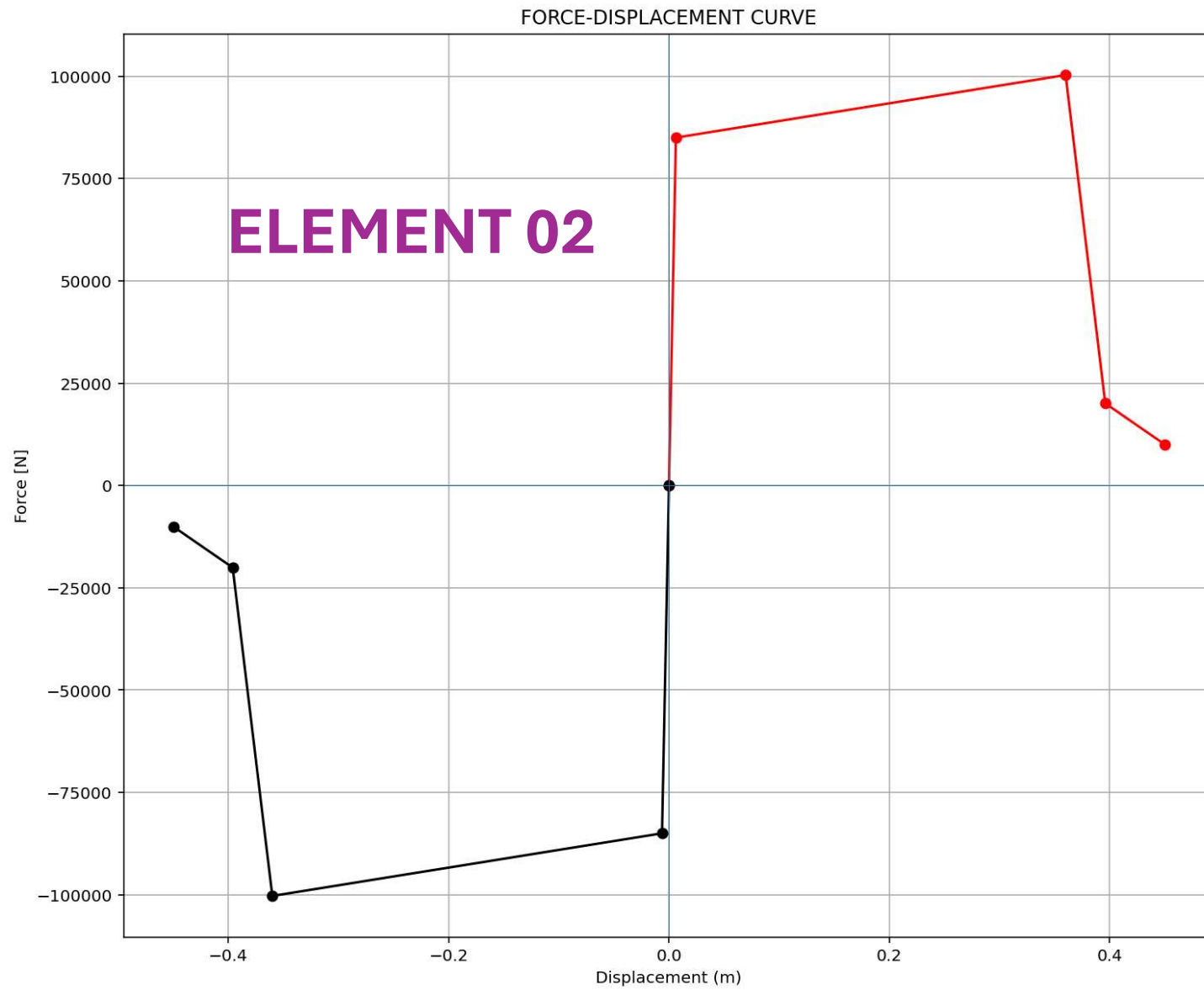
Debugger

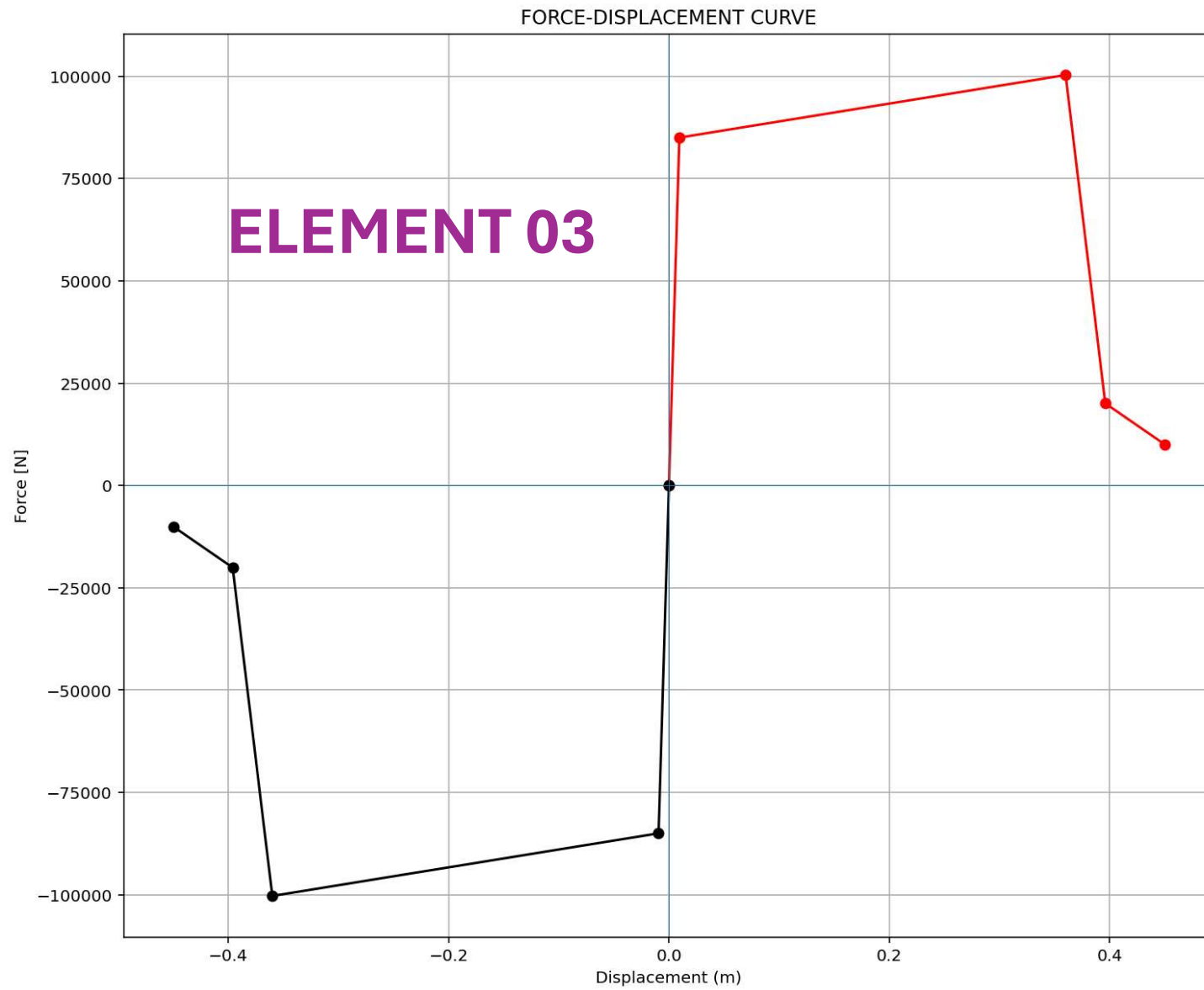
History

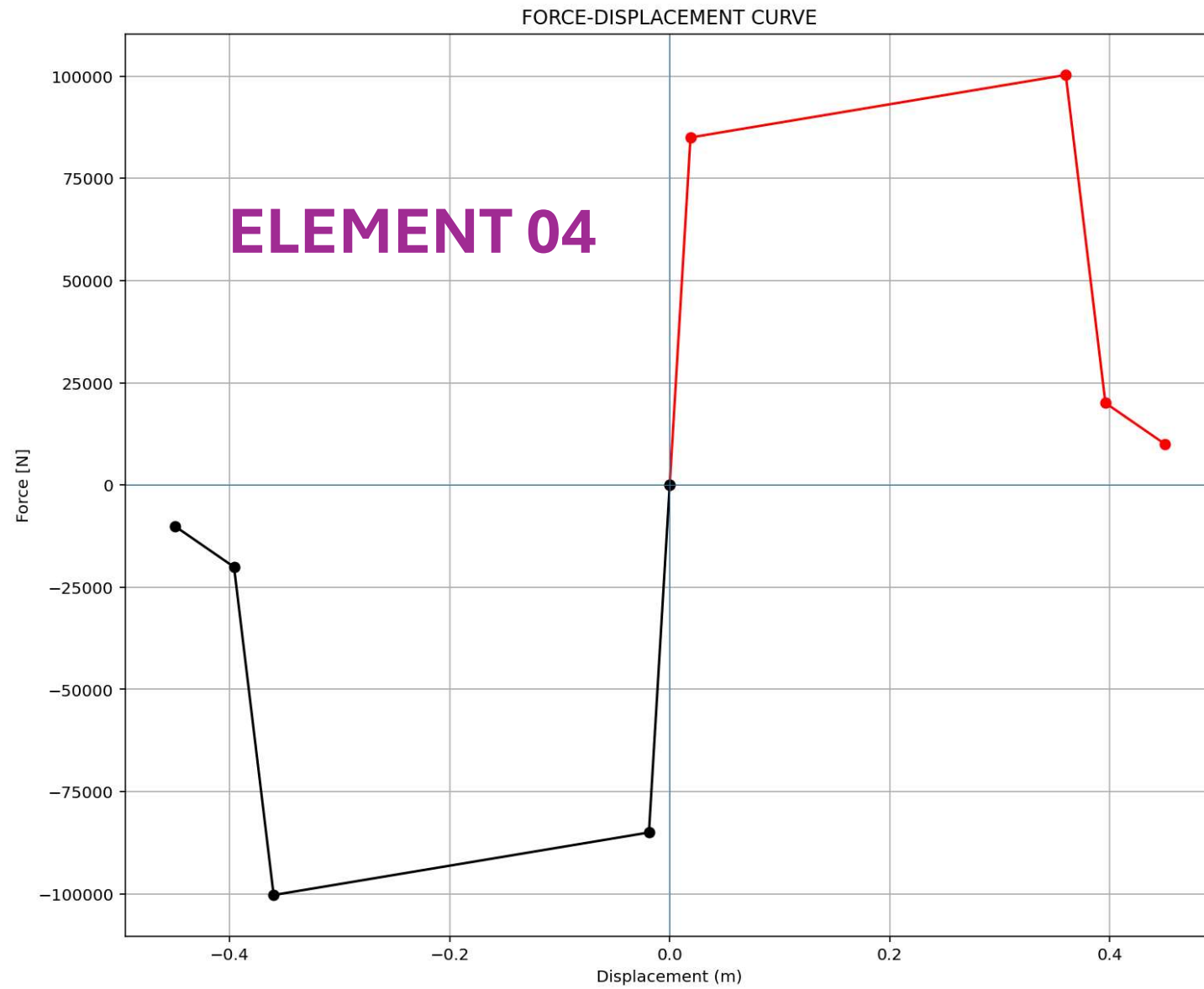
Files

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 54%

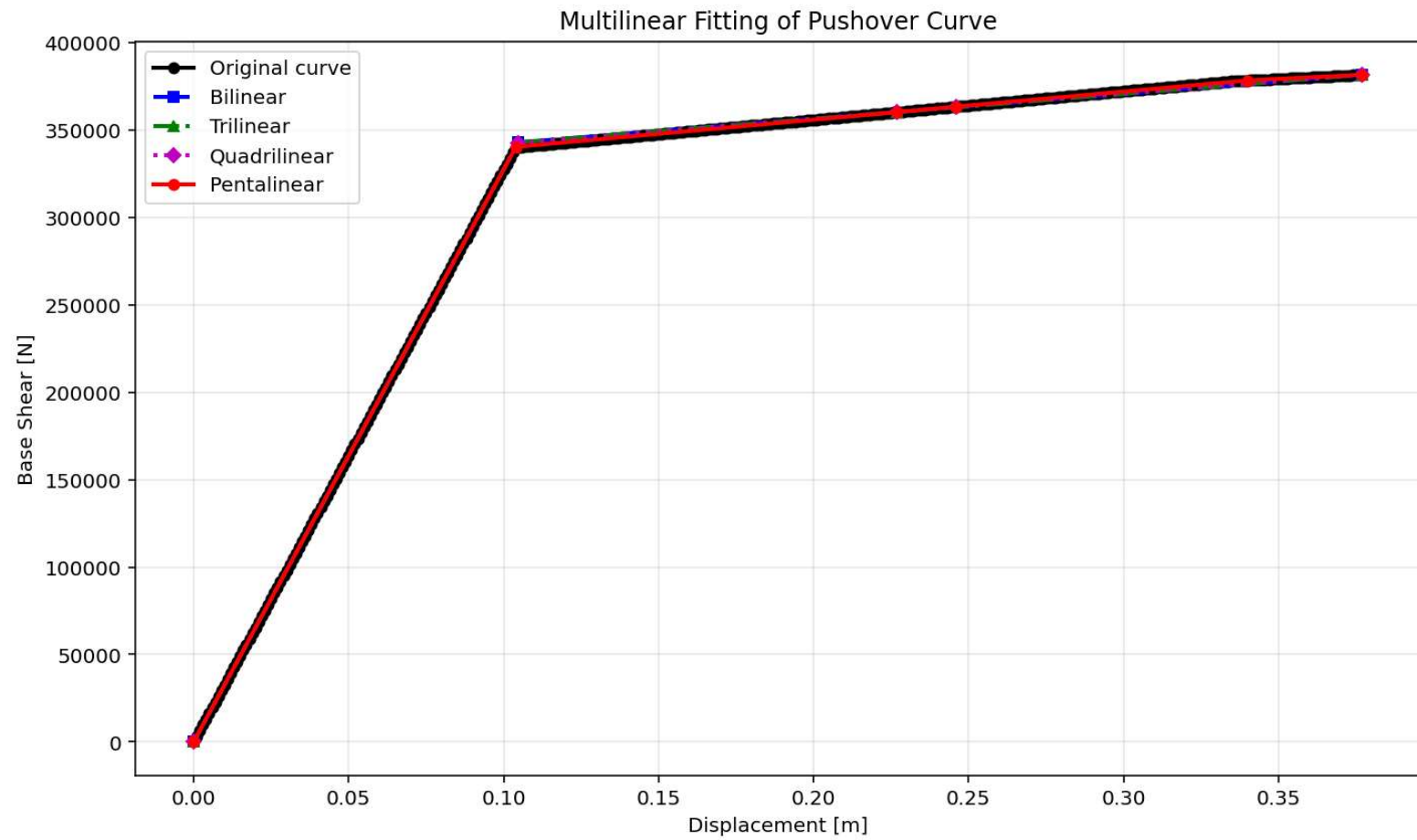


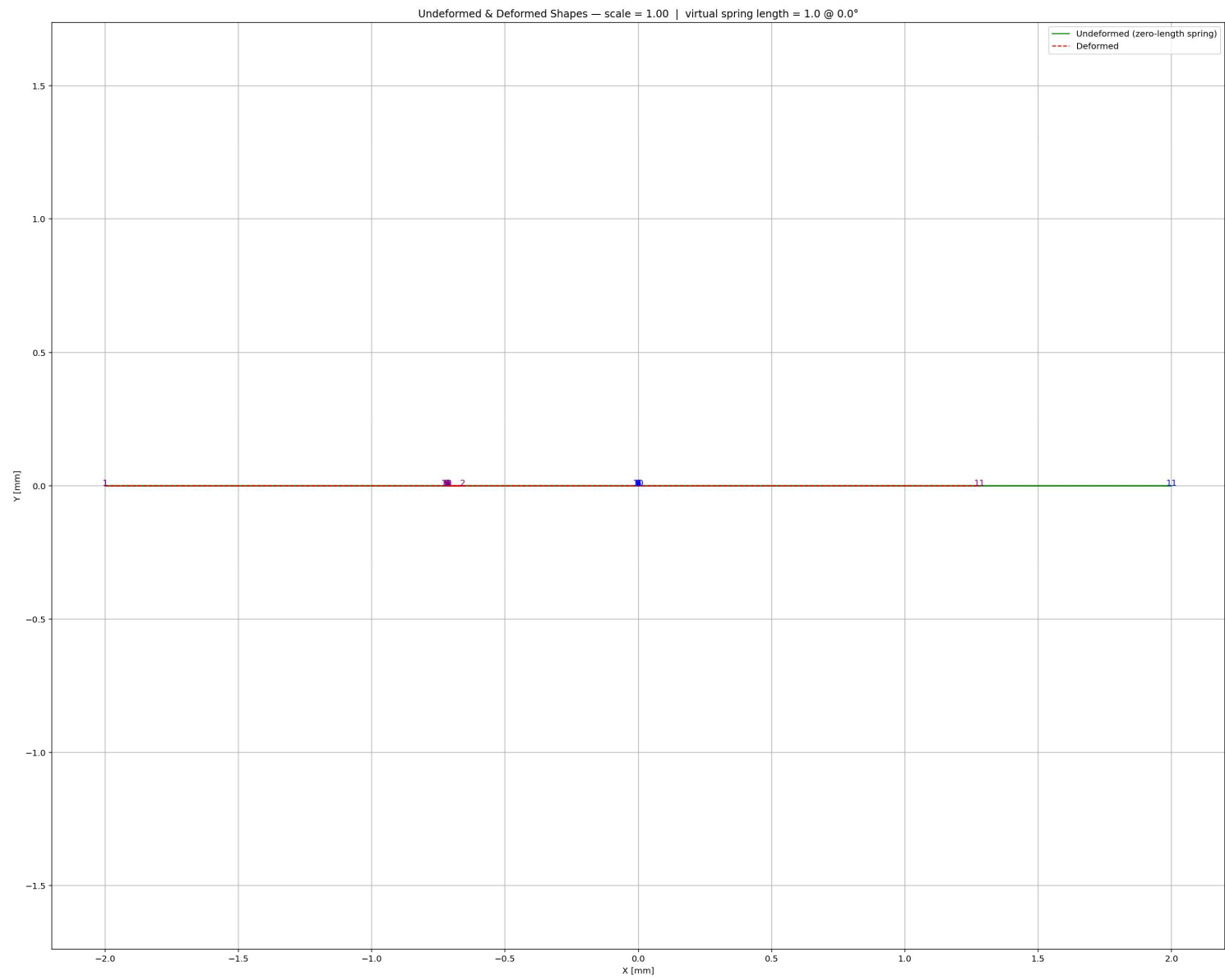






PUSHOVER ANALYSIS



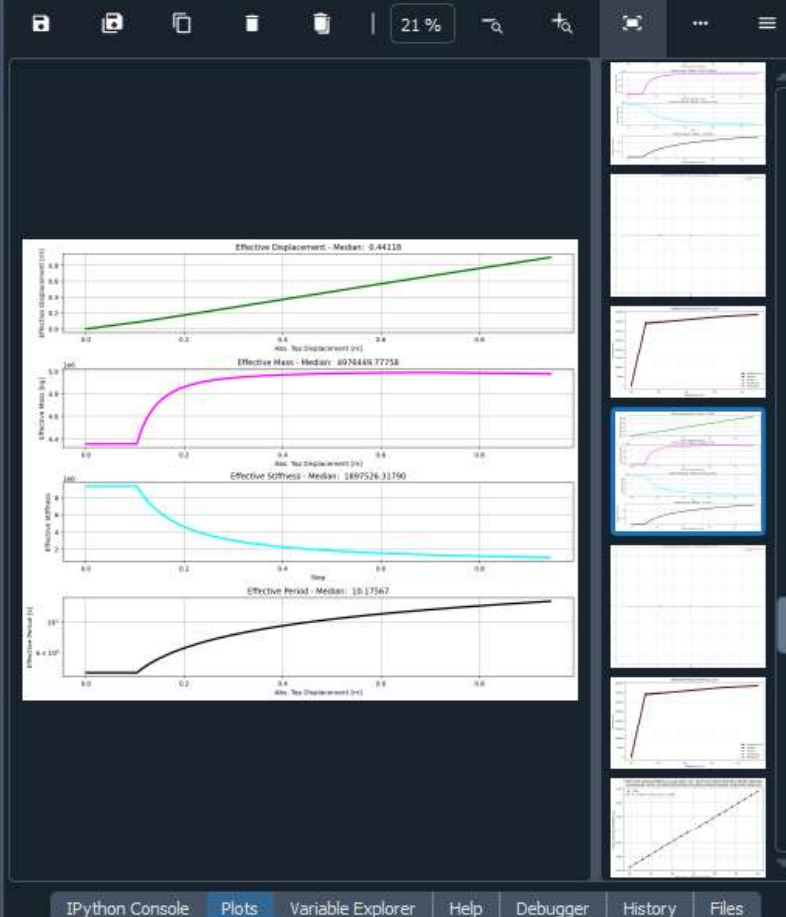


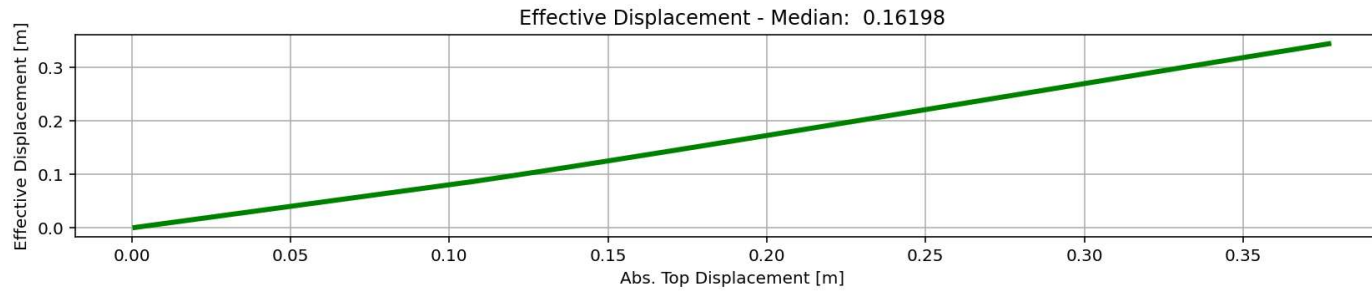
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...DOF+(COMPUTE_EFFECTIVE_PROPERTIES_FUN_PUSHOVER.py

DUCT_SENSITIVITY_PUSHOVER_MDOF.py X COMPUTE_EFFECTIVE...ES_FUN_PUSHOVER.py X

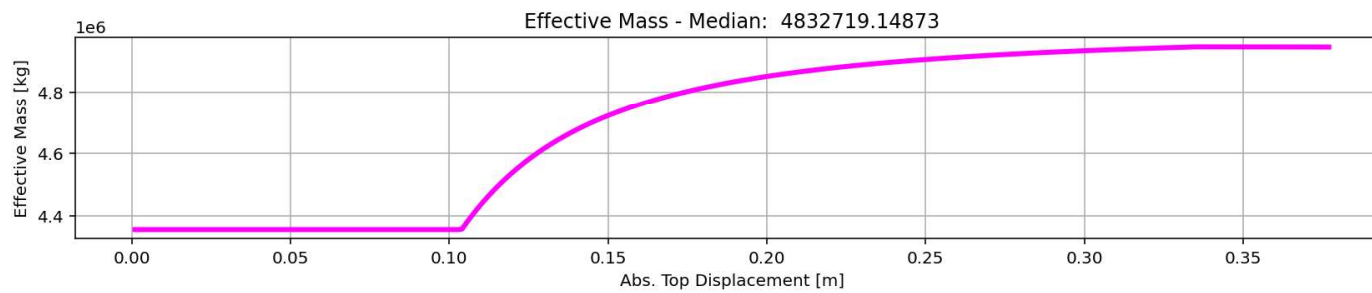
```
1  """ EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANAL
2  # Change MDOF to SDOF System with Displacement Based Design Concept
3  # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
4
5  This script implements a displacement-based pushover transformation,
6  converting a multi-degree-of-freedom (MDOF) system into an equivalent
7  single-degree-of-freedom (SDOF) system for seismic assessment.
8  It calculates effective modal properties-displacement, mass, and
9  stiffness-by weighting element forces and nodal displacements according
10 to a presumed deformed shape.
11 The derived effective period provides a simplified dynamic characteristic
12 for performance-based engineering. The visualizations effectively track the
13 evolution of these equivalent parameters throughout the nonlinear analysis steps.
14 """
15
16 #-----
17 displacement_X_1 = np.array(list(node_displacements.values())[1]) # DOF 02
18 displacement_X_2 = np.array(list(node_displacements.values())[2]) # DOF 03
19 displacement_X_3 = np.array(list(node_displacements.values())[3]) # DOF 04
20 displacement_X_4 = np.array(list(node_displacements.values())[4]) # DOF 05
21 displacement_X_5 = np.array(list(node_displacements.values())[5]) # DOF 06
22 displacement_X_6 = np.array(list(node_displacements.values())[6]) # DOF 07
23 displacement_X_7 = np.array(list(node_displacements.values())[7]) # DOF 08
24 displacement_X_8 = np.array(list(node_displacements.values())[8]) # DOF 09
25 displacement_X_9 = np.array(list(node_displacements.values())[9]) # DOF 10
26 displacement_X_10 = np.array(list(node_displacements.values())[10]) # DOF 11
27
28 ele_force_01 = np.array(list(ele_force.values())[0]) # ELEMENT 01
29 ele_force_02 = np.array(list(ele_force.values())[1]) # ELEMENT 02
30 ele_force_03 = np.array(list(ele_force.values())[2]) # ELEMENT 03
31 ele_force_04 = np.array(list(ele_force.values())[3]) # ELEMENT 04
32 ele_force_05 = np.array(list(ele_force.values())[4]) # ELEMENT 05
33 ele_force_06 = np.array(list(ele_force.values())[5]) # ELEMENT 06
34 ele_force_07 = np.array(list(ele_force.values())[6]) # ELEMENT 07
35 ele_force_08 = np.array(list(ele_force.values())[7]) # ELEMENT 08
```

...FLOORS_04_COLUMNS_8_ANA\DUCT_SENSITIVITY_PUSHOVER_MDOF+

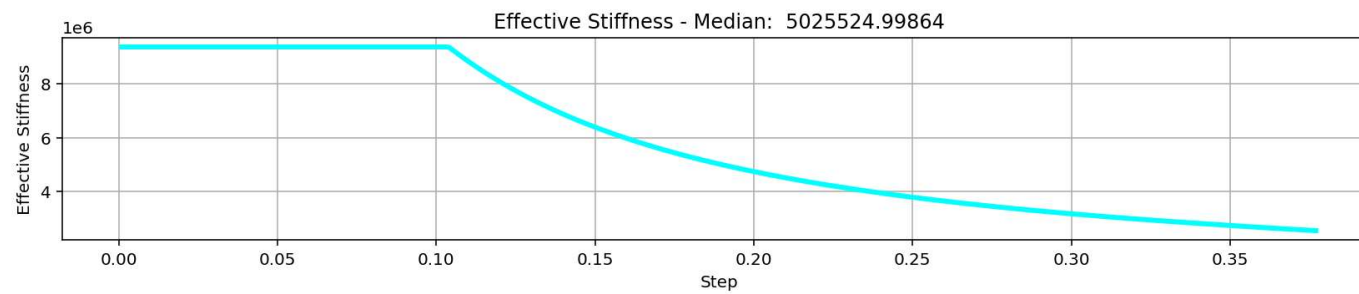




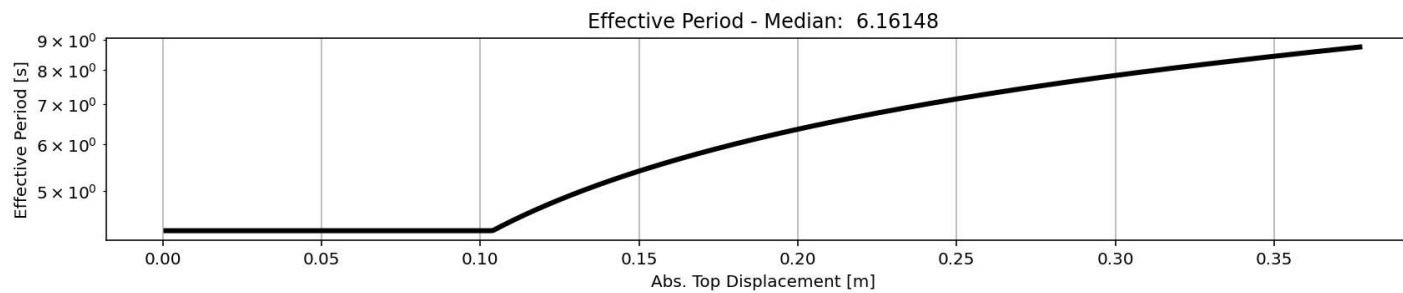
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$



$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$



$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$



$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

BILINEAR CURVE

+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[1.05586496e-01 3.45555806e+05]

[9.44000000e-01 3.86740624e+05]]

+=====+

+-----+

Structure Elastic Stiffness : 3272727.27

Structure Plastic Stiffness : 409682.86

Structure Tangent Stiffness : [3272727.27 49122.32]

Structure Ductility Ratio : 8.94

Structure Over Strength Factor: 1.12

+-----+

TRILINEAR CURVE

+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[1.04512702e-01 3.42041571e+05]

[5.67000000e-01 3.70124071e+05]

[9.44000000e-01 3.86740624e+05]]

+=====+

+-----+

Structure Elastic Stiffness : 3272727.27

Structure Plastic Stiffness : 409682.86

Structure Tangent Stiffness : [3272727.27 60720.59 44075.74]

Structure Ductility Ratio : 9.03

Structure Over Strength Factor: 1.13

+-----+

QUADRILINEAR CURVE

+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[1.04262557e-01 3.41222915e+05]

[5.67000000e-01 3.70124071e+05]

[6.14000000e-01 3.73181289e+05]

[9.44000000e-01 3.86740624e+05]]

+=====+

+-----+

Structure Elastic Stiffness : 3272727.27

Structure Plastic Stiffness : 409682.86

Structure Tangent Stiffness : [3272727.27 62456.92 65047.18 41088.89]

Structure Ductility Ratio : 9.05

Structure Over Strength Factor: 1.13

+-----+

PENTILINEAR CURVE

+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[1.04126239e-01 3.40776781e+05]

[5.67000000e-01 3.70124071e+05]

[6.14000000e-01 3.73181289e+05]

[8.50000000e-01 3.83491914e+05]

[9.44000000e-01 3.86740624e+05]]

+=====+

+-----+

Structure Elastic Stiffness : 3272727.27

Structure Plastic Stiffness : 409682.86

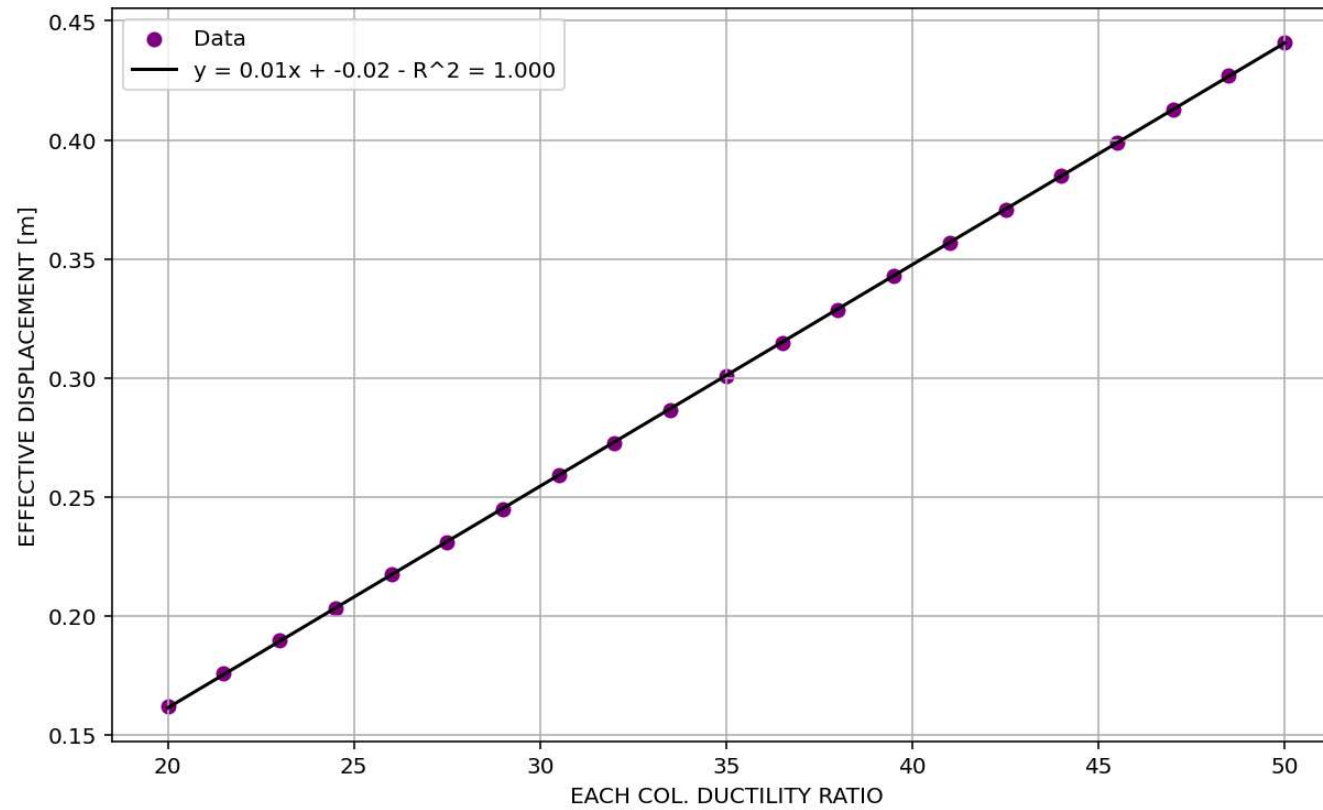
Structure Tangent Stiffness : [3272727.27 63402.36 65047.18 43689.09 34560.74]

Structure Ductility Ratio : 9.07

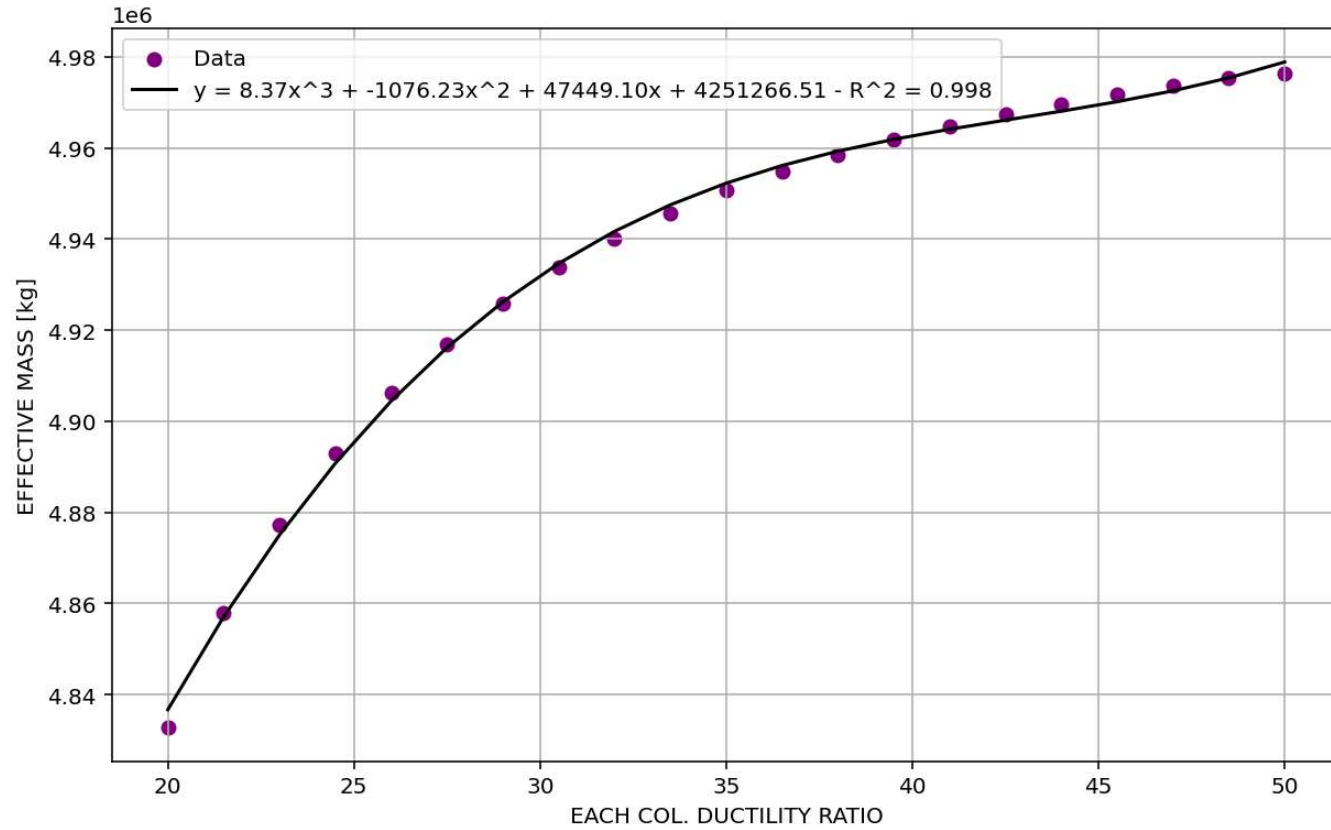
Structure Over Strength Factor: 1.13

+-----+

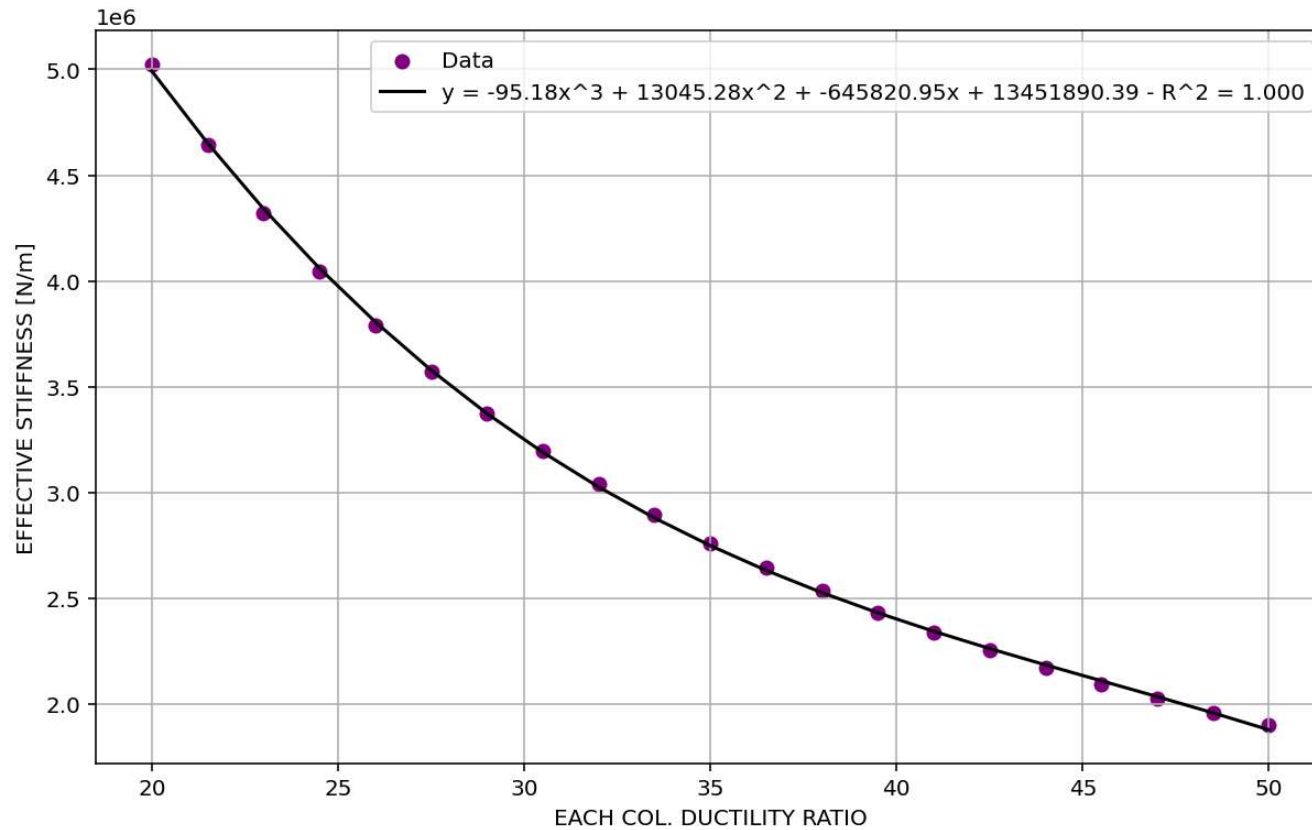
EFFECTIVE DISPLACEMENT [m] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS



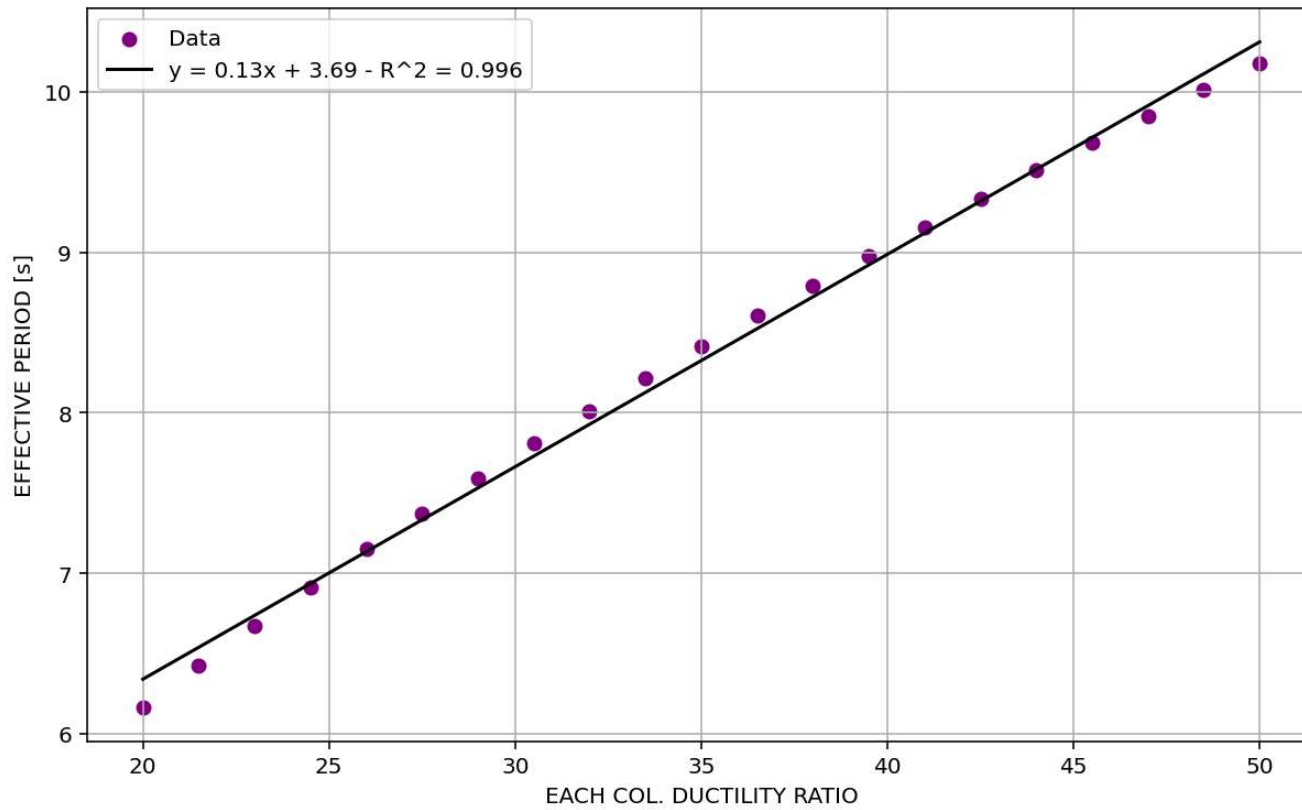
EFFECTIVE MASS [kg] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS



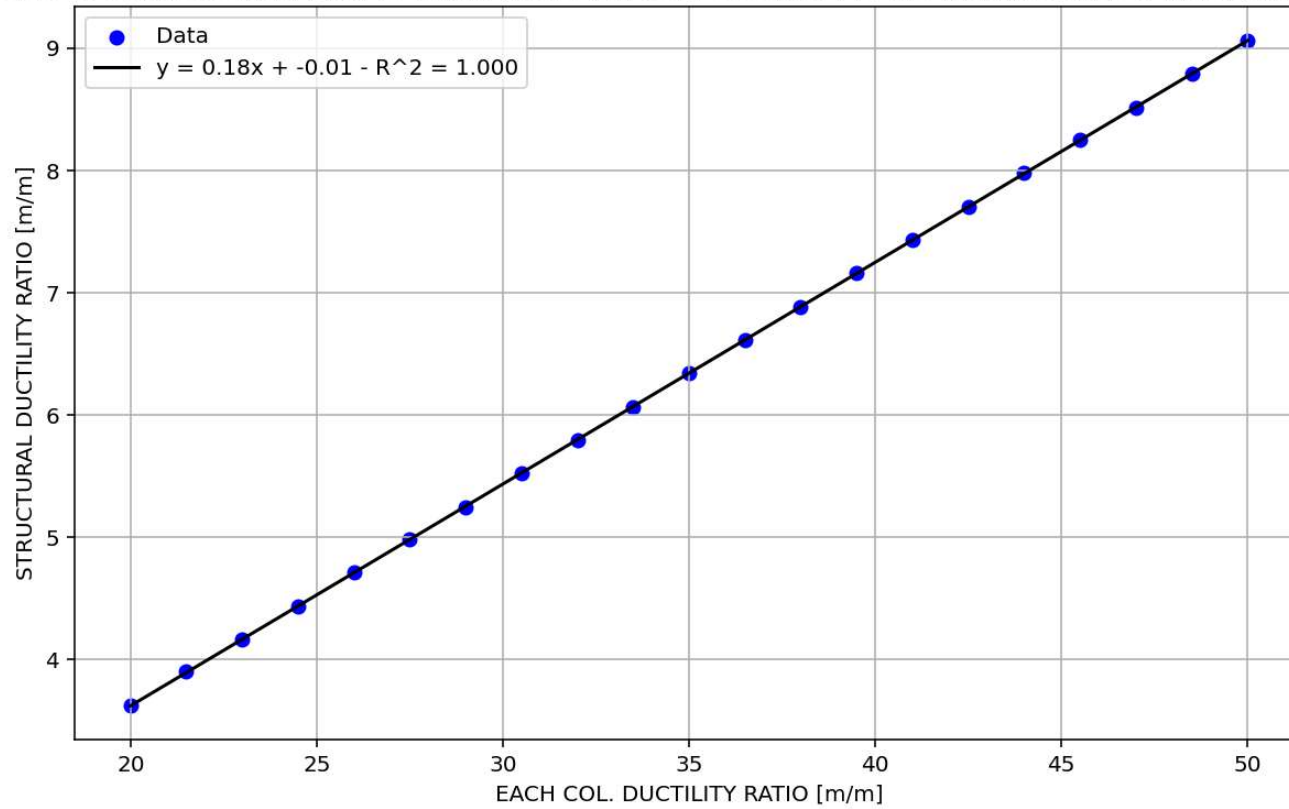
EFFECTIVE STIFFNESS [N/m] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS



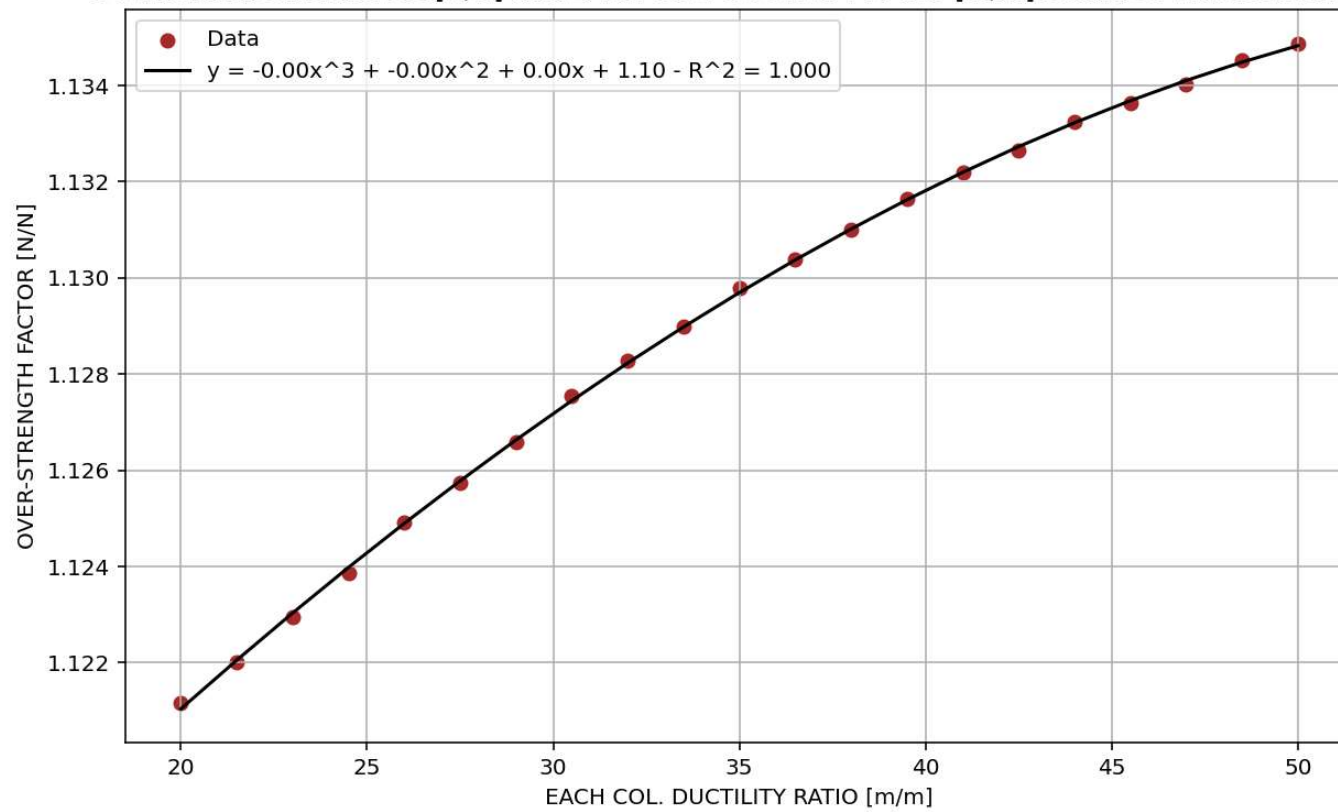
EFFECTIVE PERIOD [s] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS



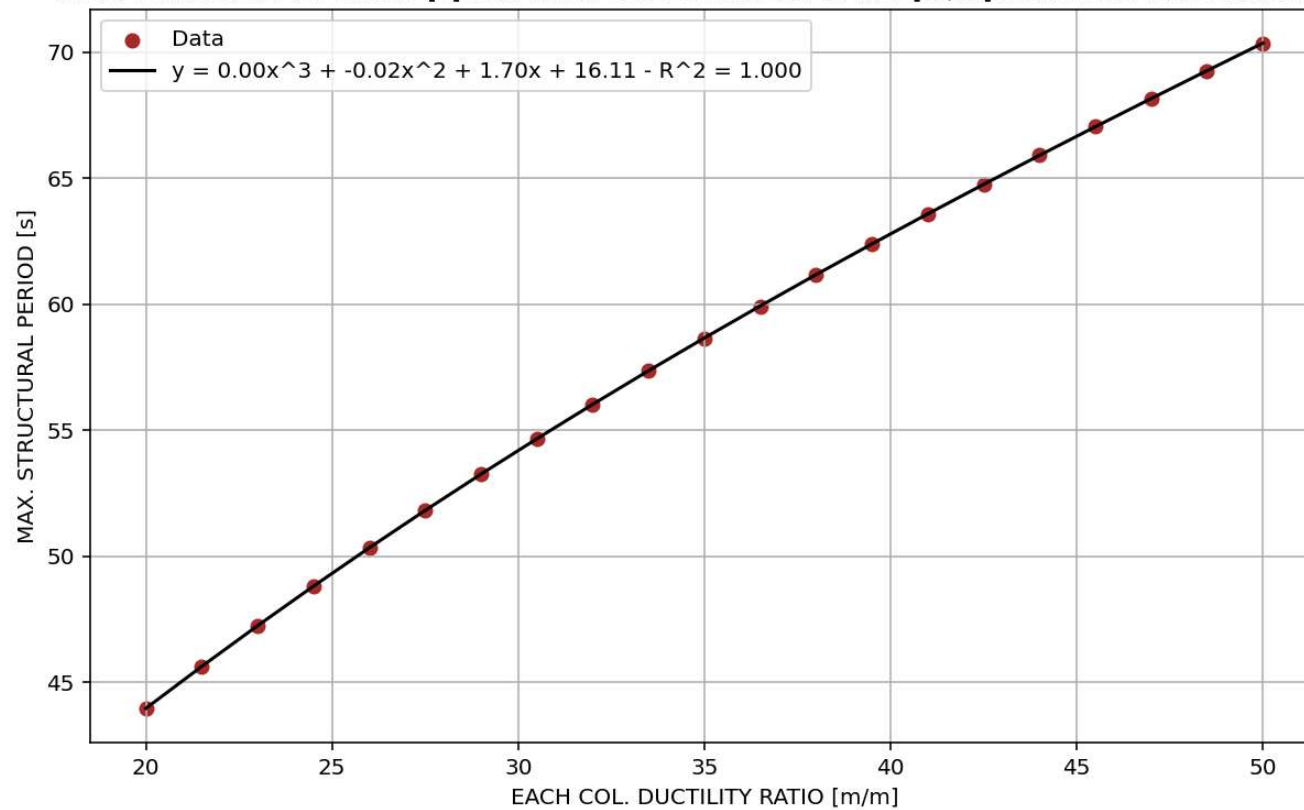
STRUCTURAL DUCTILITY RATIO [m/m] and EACH COL. DUCTILITY RATIO [m/m] DURING PERIOD ANALYSIS

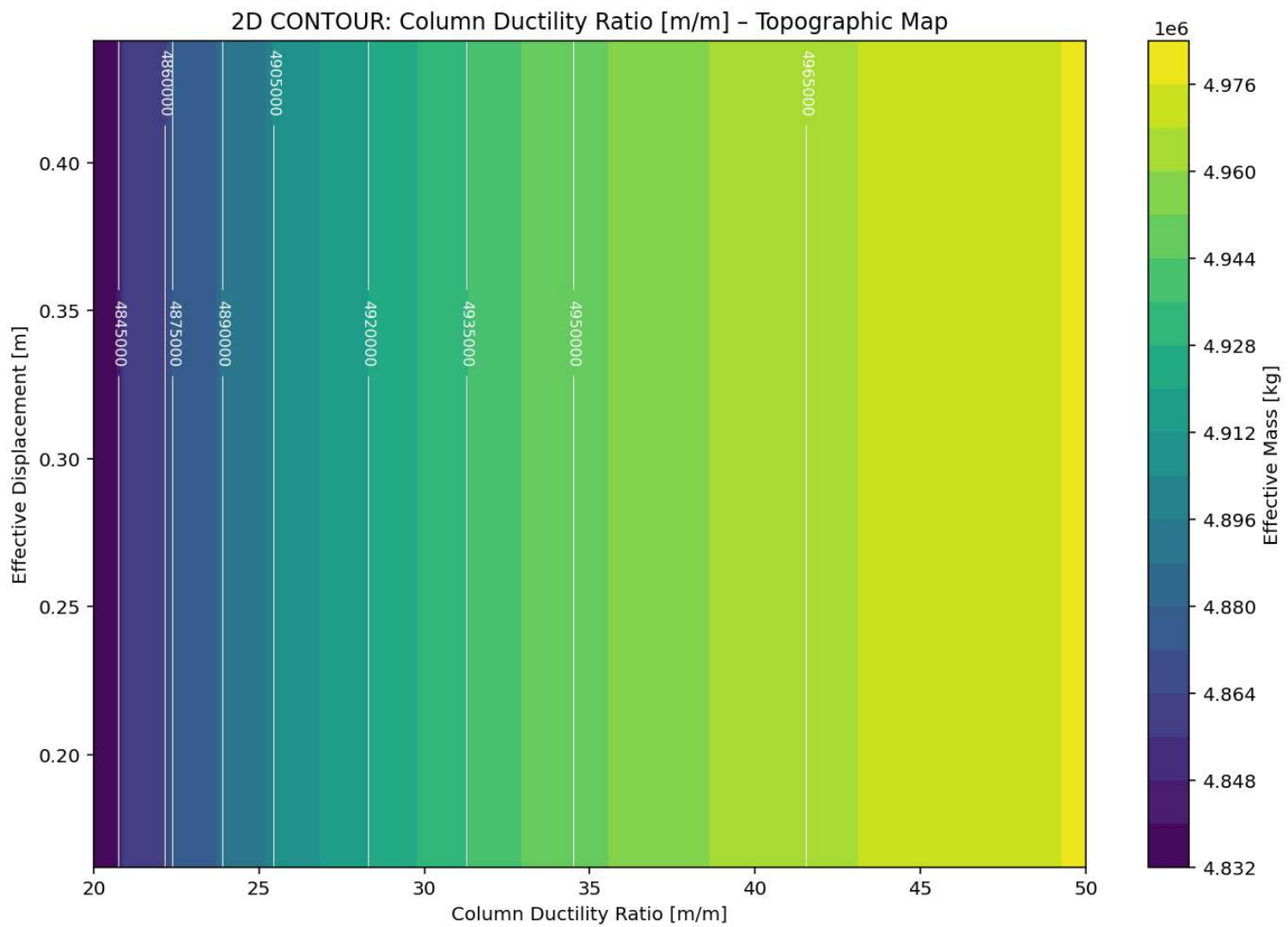


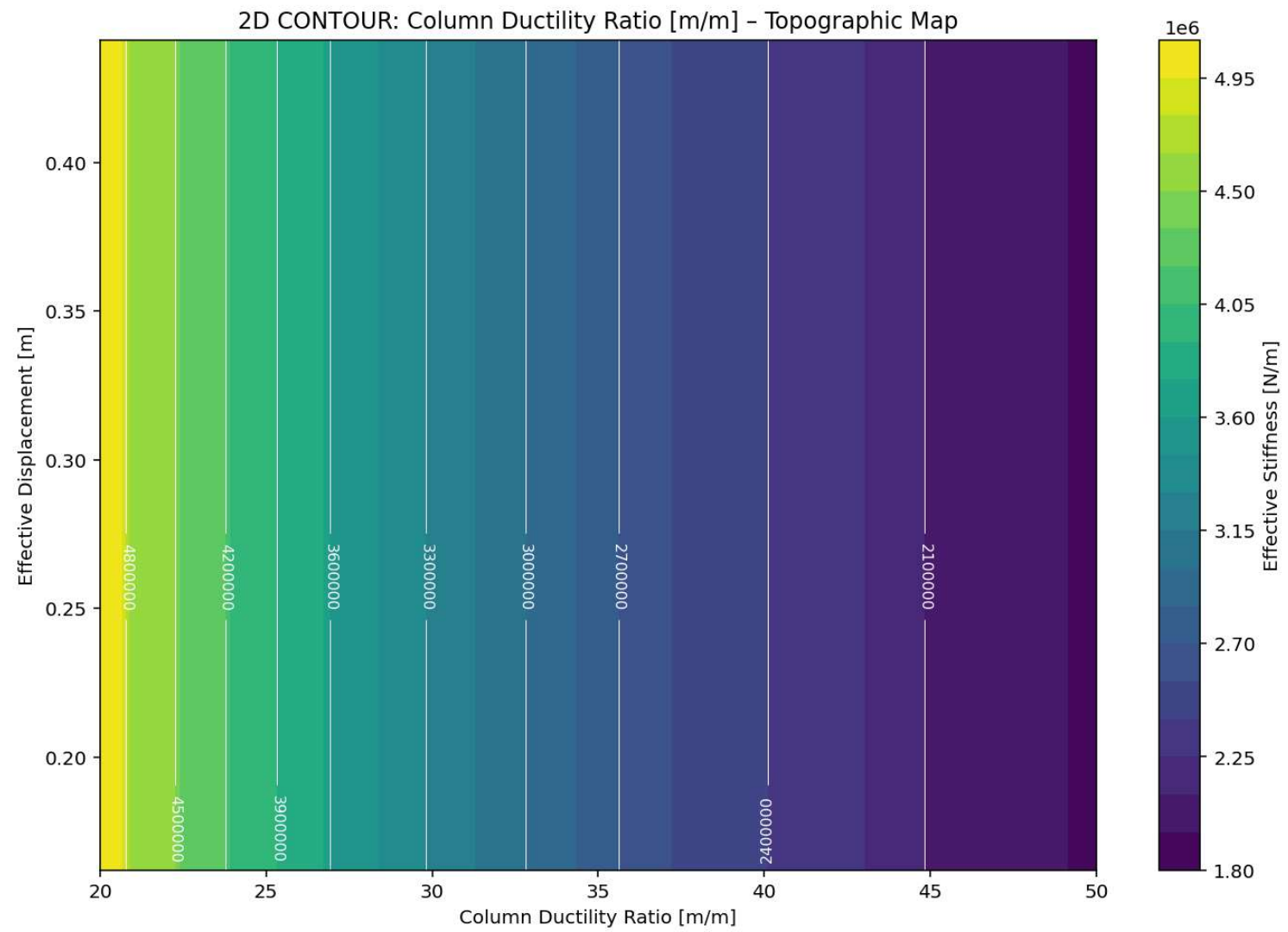
OVER-STRENGTH FACTOR [N/N] and EACH COL. DUCTILITY RATIO [m/m] DURING PERIOD ANALYSIS

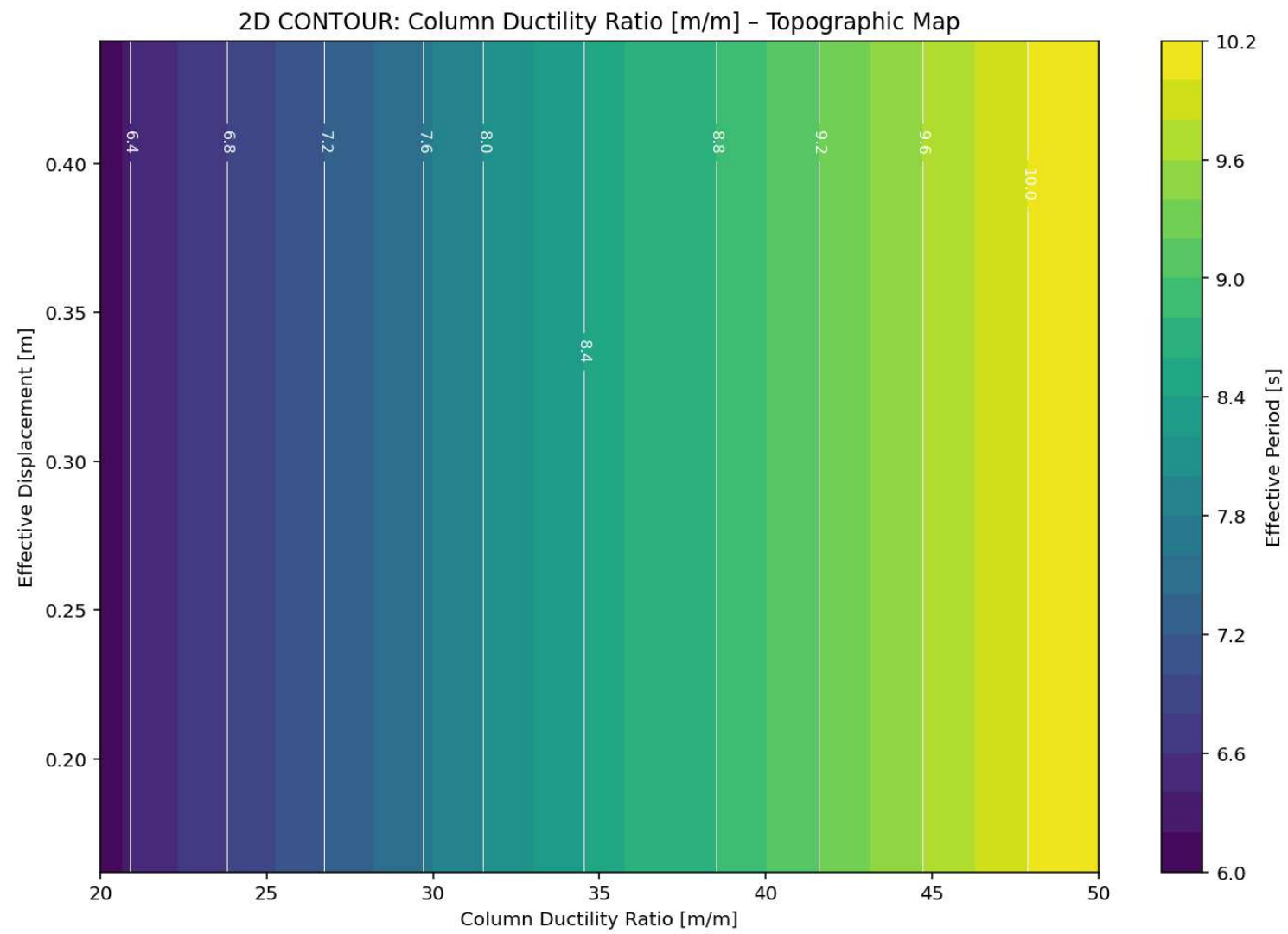


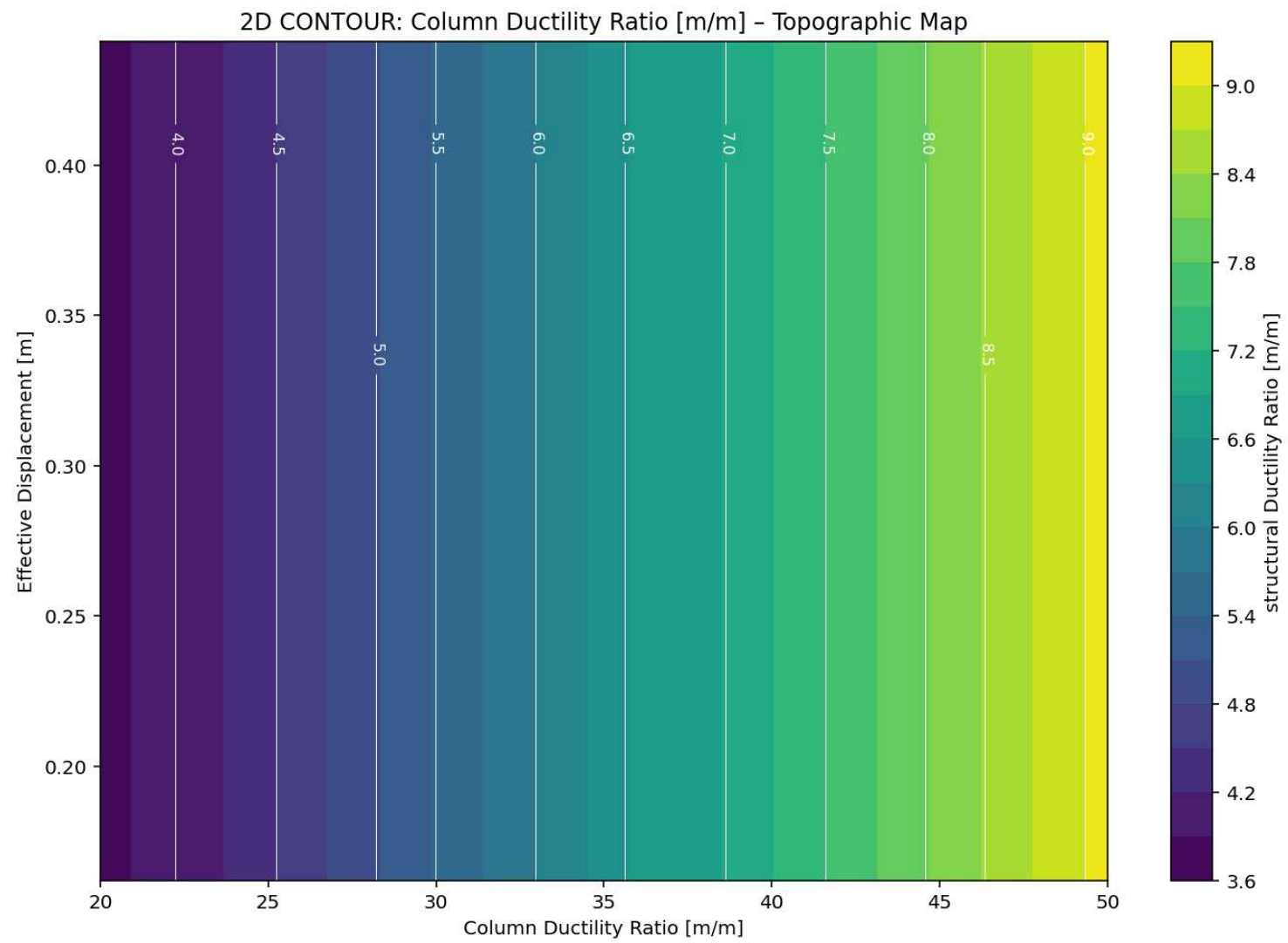
MAX. STRUCTURAL PERIOD [s] and EACH COL. DUCTILITY RATIO [m/m] DURING PERIOD ANALYSIS

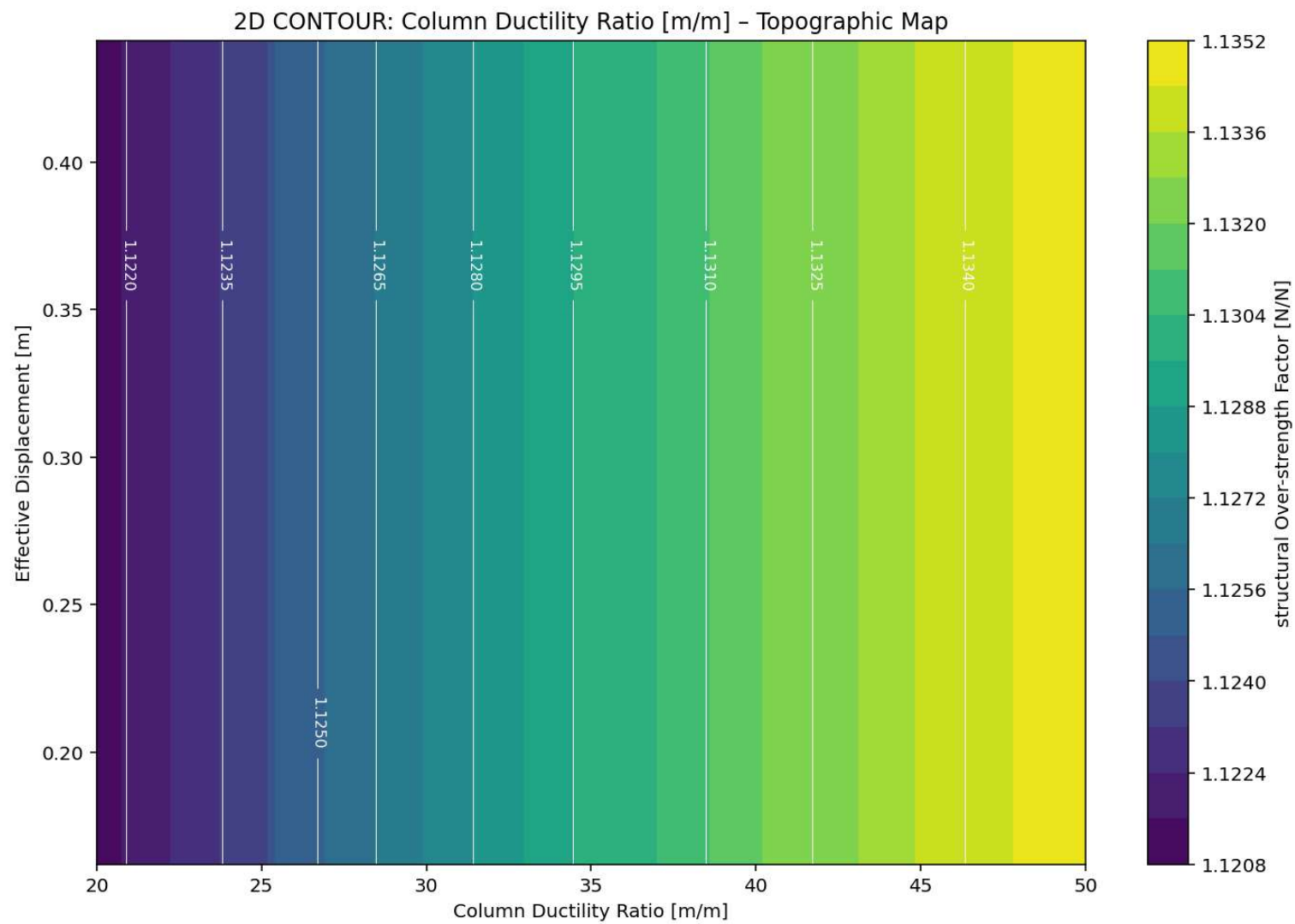


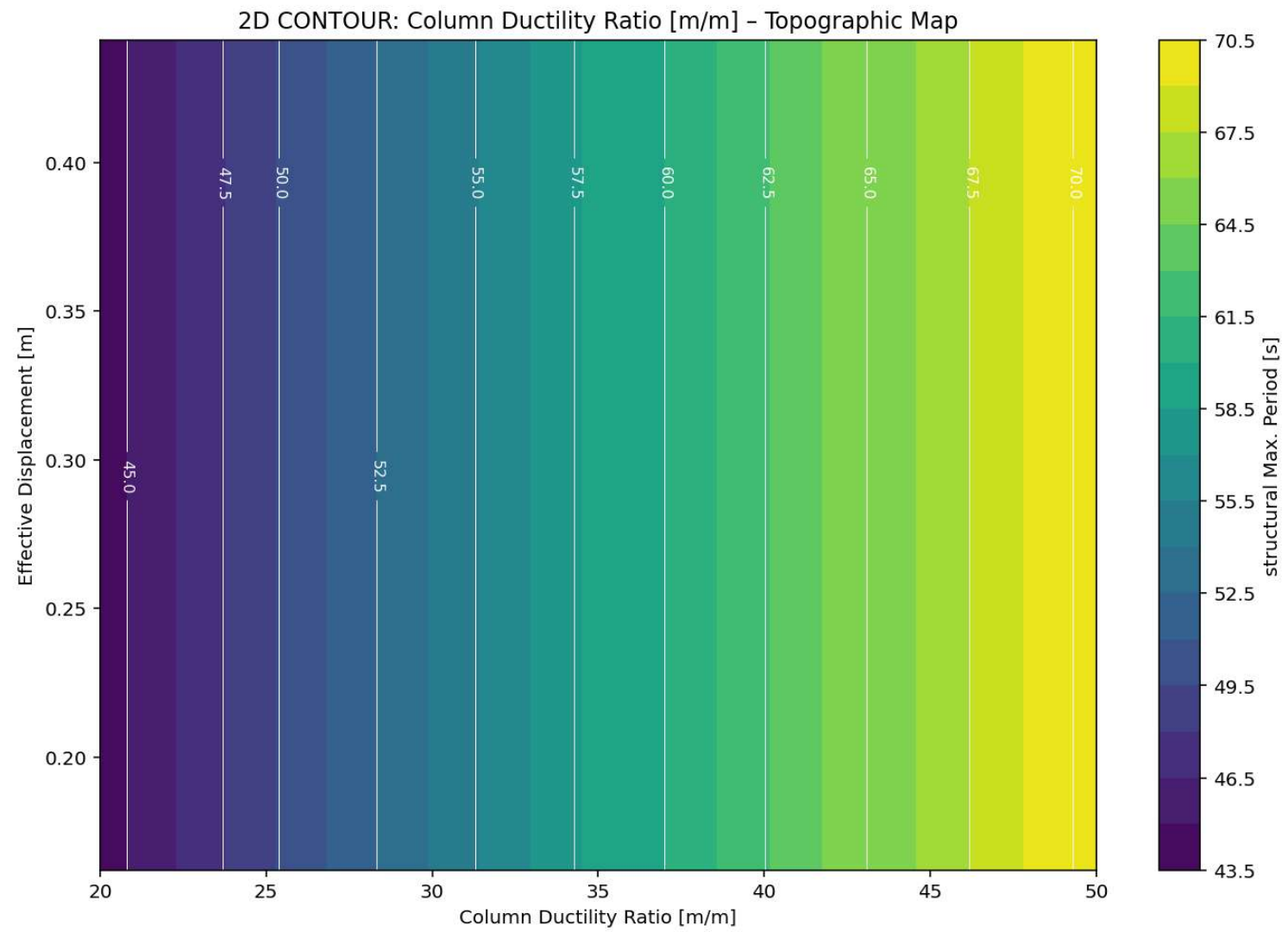


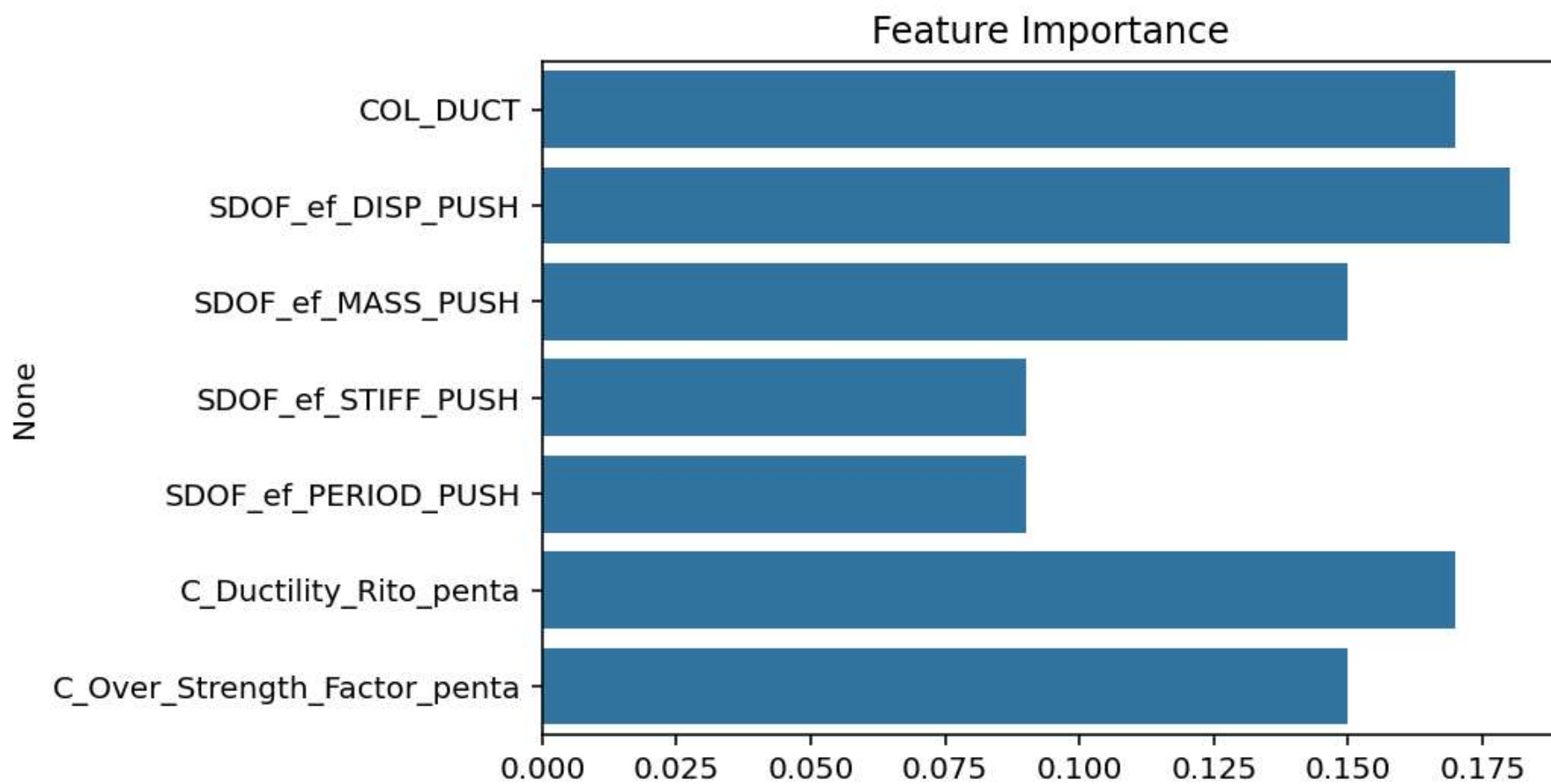




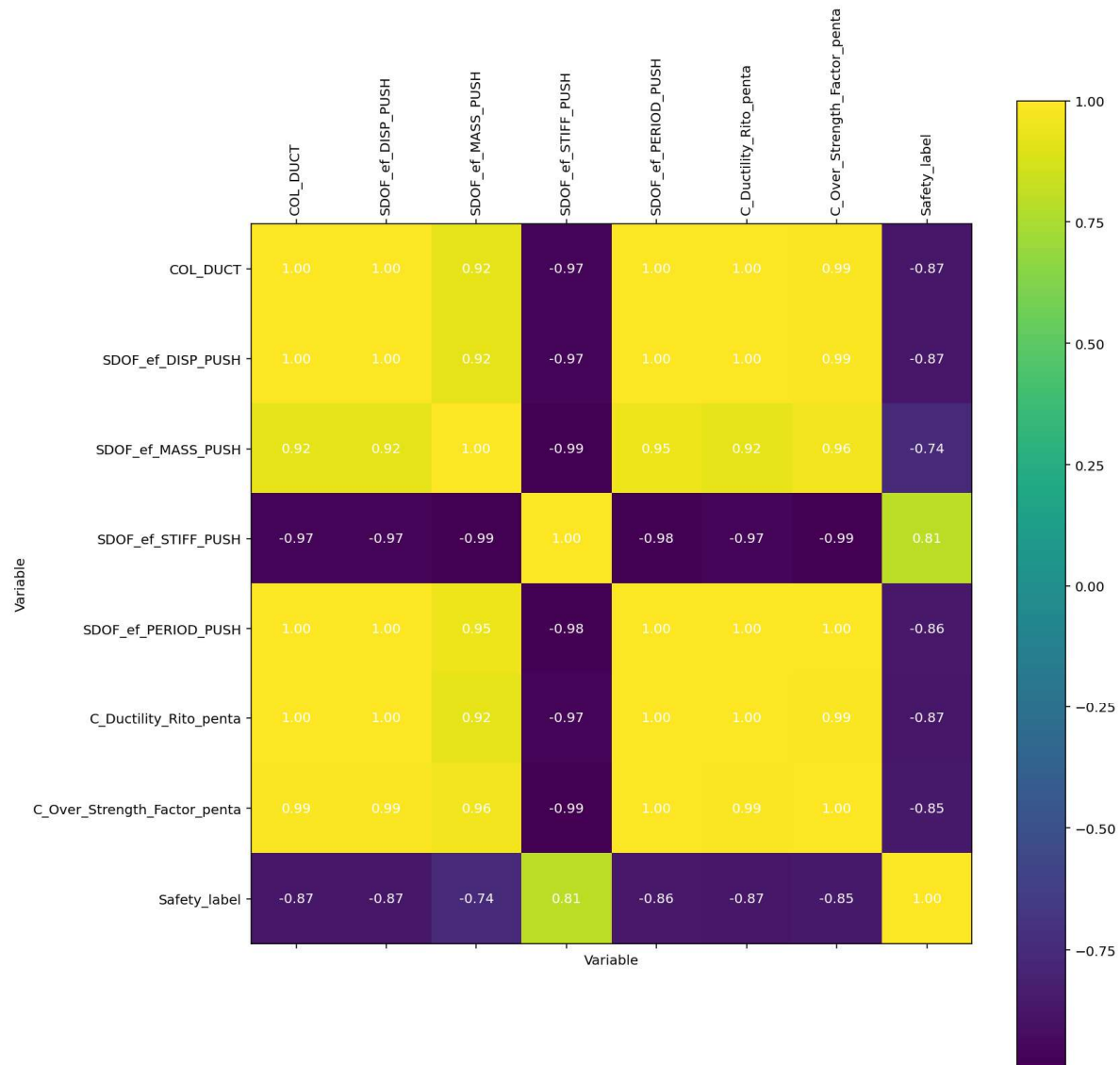


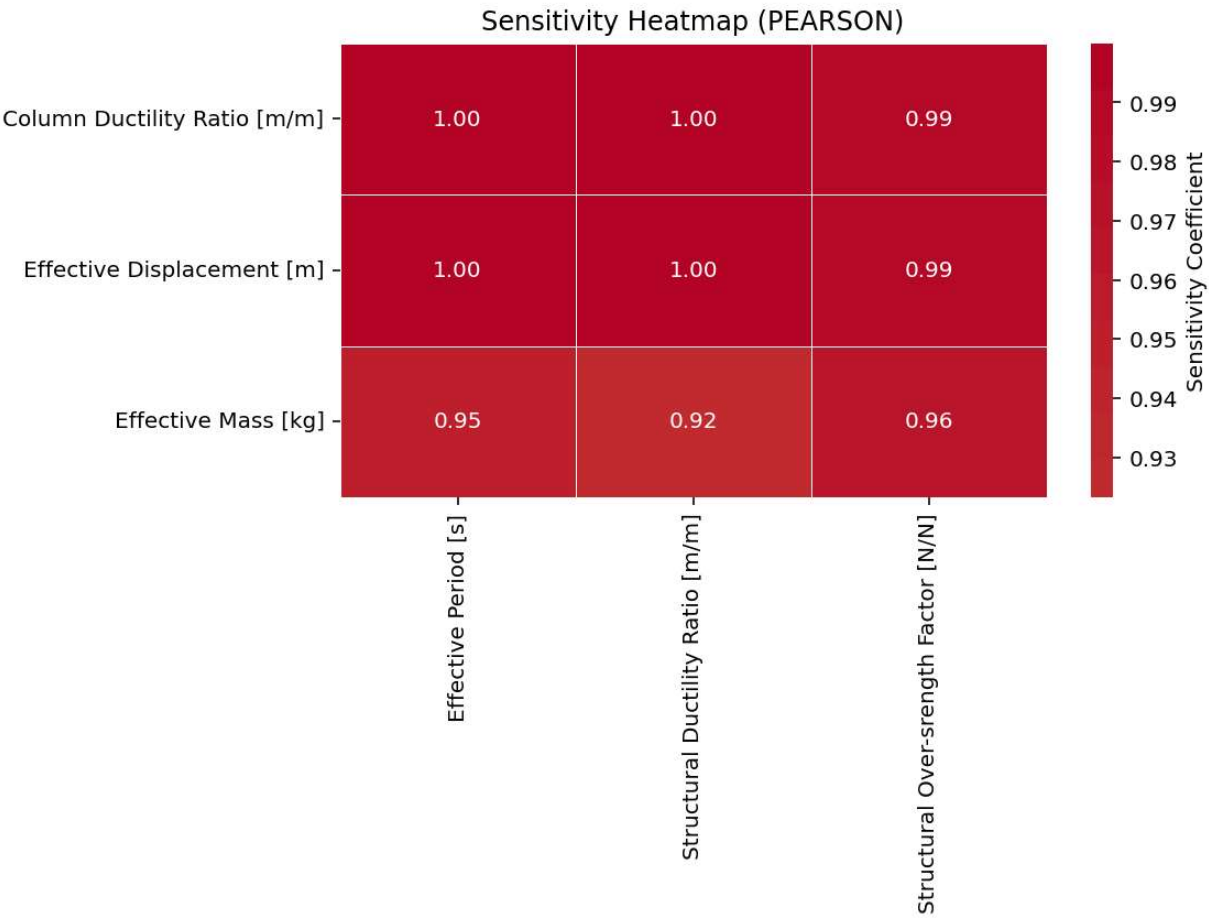


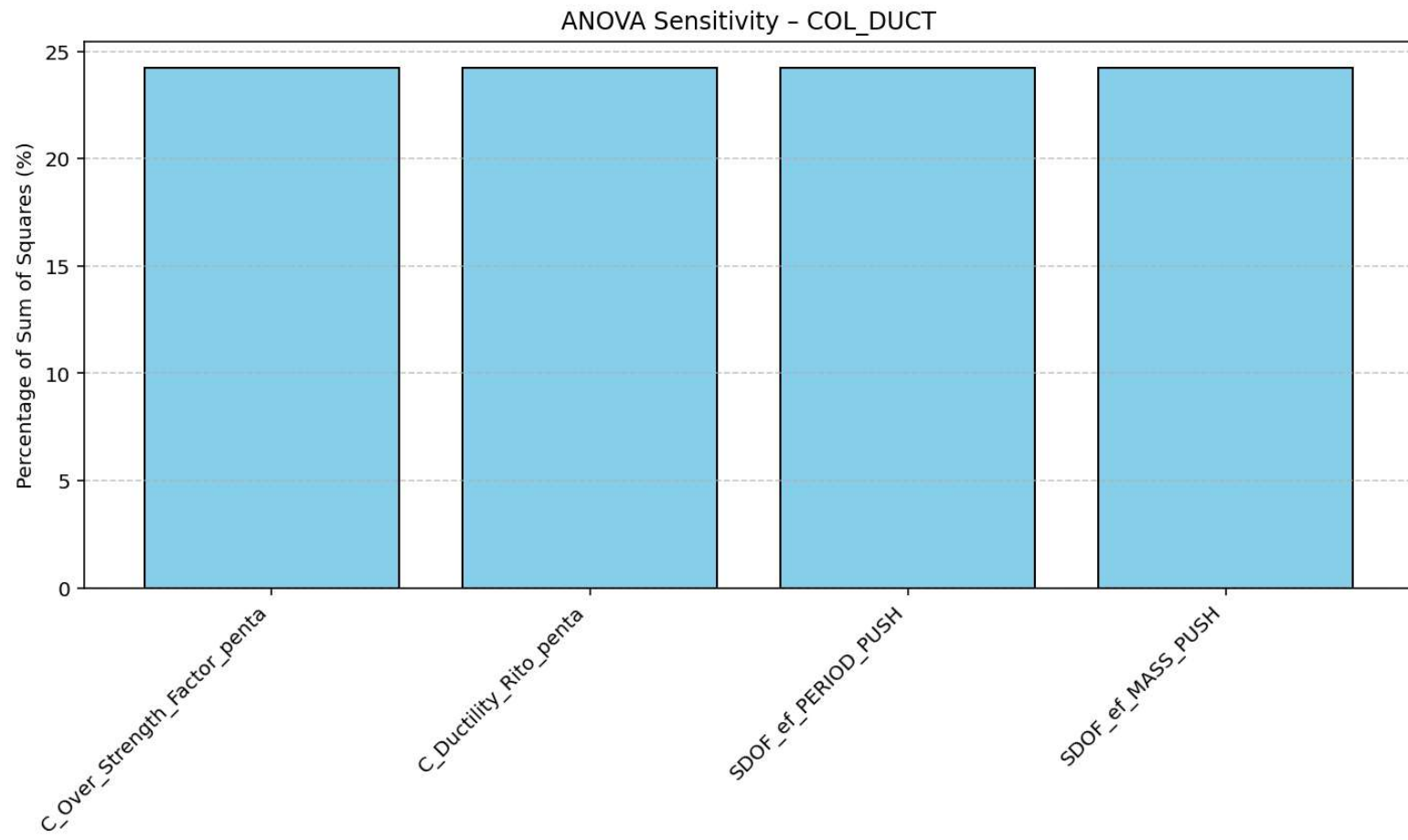


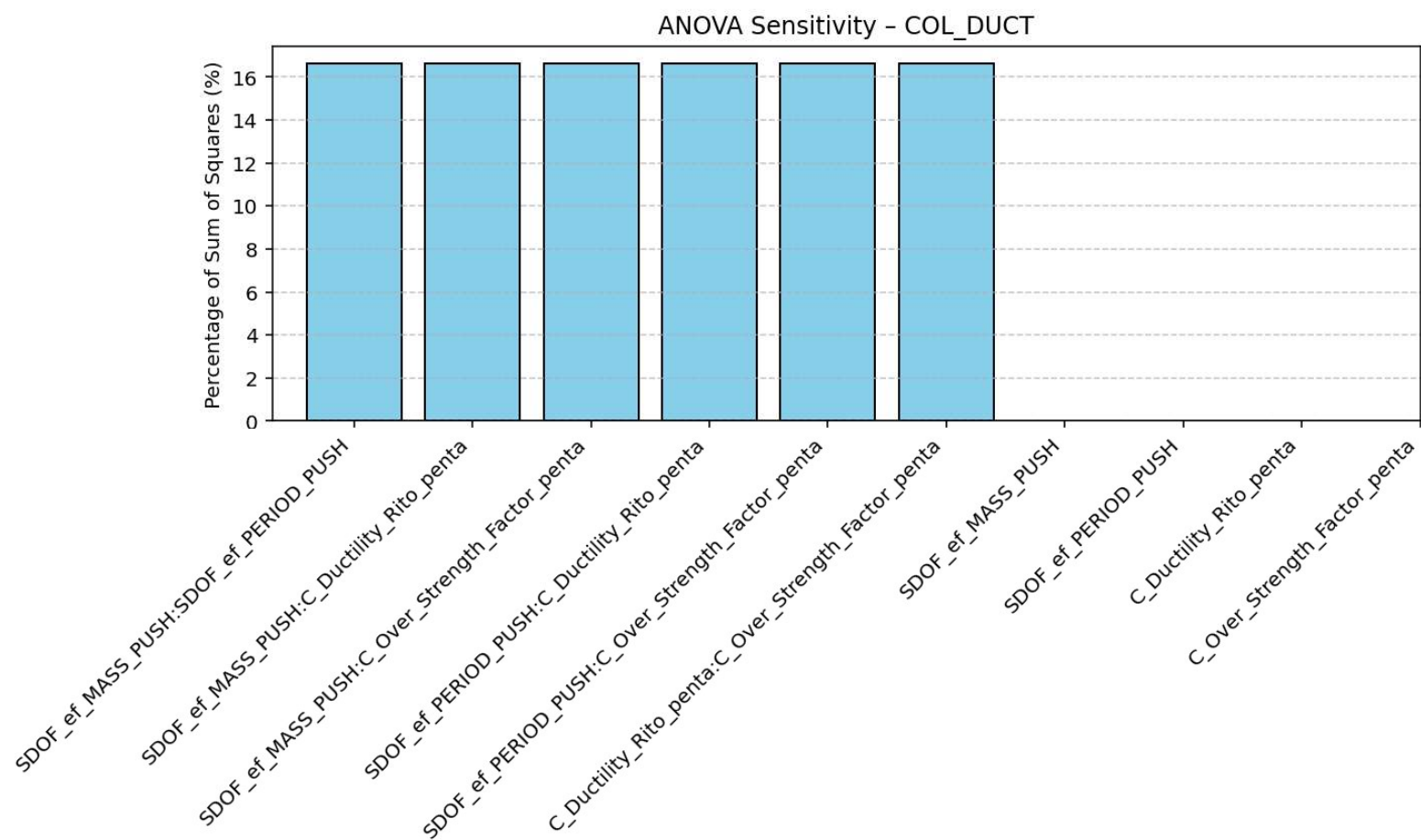


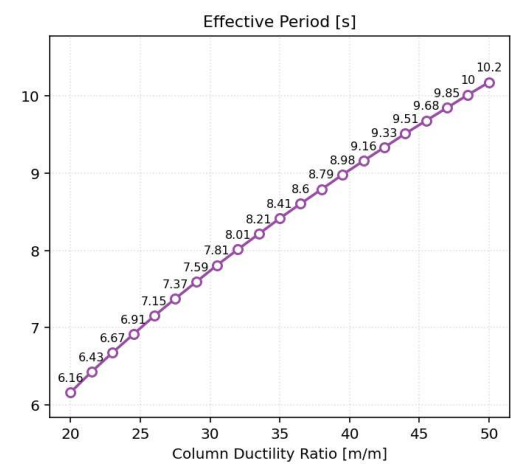
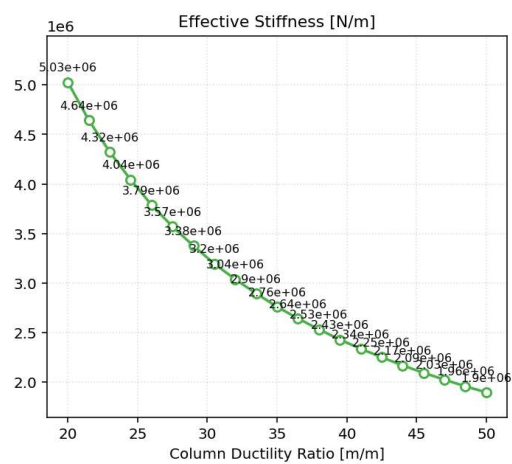
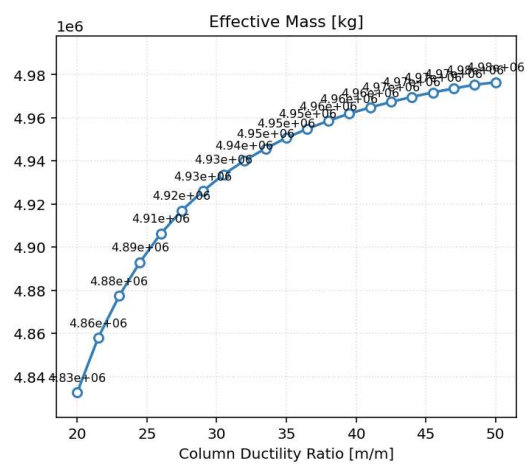
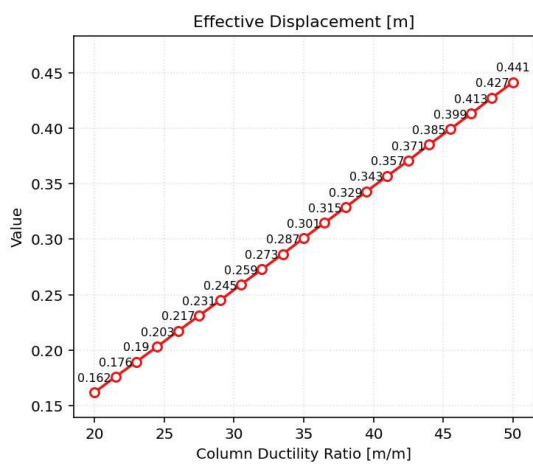
Correlation Heatmap











Statistical Metrics vs Column Ductility Ratio

